UMEÅ UNIVERSITY

Monthly Electricity Consumption Forecasting Using an Explainable AI Framework

Tyra Wodén

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Spring 2026
Degree Project in Interaction Technology and Design, 30 credits
Supervisor: Esteban Guerrero Rosero
External Supervisor: Elin Eriksson
Examiner: Ola Ringdahl
Master of Science Program in Interaction Technology and Design, 300 credits

Abstract

Recent years have seen instability and increasing demands in the electricity market. Reliable electricity consumption forecasting is a helpful tool for consumers and stakeholders to manage energy consumption more efficiently. This thesis forecasts the monthly electricity consumption of individual residential households using five robust ensemble learning models. To address a gap in previous research, the interpretability of the monthly residential forecasts is enhanced using insights produced by the explainable artificial intelligence (XAI) method SHAP. A diverse set of features, including historical consumption, household attributes, and weather data, is investigated for their contributions to model output. Results show that the ensemble models are able to forecast the consumption with around 15% prediction error, although performance is highly impacted by extreme values for some households. The XAI analysis revealed recent consumption to be the most determining factor for the forecasts, while many external features have a complex or weak relationship to model output. Furthering the research on interpretable forecasts is important, so that end-users are able to trust the forecasts and subsequently manage their energy consumption more efficiently.

Keywords: household electricity consumption, monthly electricity consumption, decision tree ensemble learning, explainable artificial intelligence (XAI), SHapley Additive exPlanations (SHAP)

Checklist for final touches:

- Clean up references
- (Center page content?)

Acknowledgements

Contents

List of Abbreviations

AI Artificial Intelligence

ML Machine Learning

MAPE Mean Absolute Percentage Error

RMSE Root Mean Square Error

NRMSE Normalized Root Mean Square Error

MAE Mean Absolute Error

XAI Explainable Artificial Intelligence

SHAP SHapley Additive exPlanations

RF Random Forest

GBM Gradient Boosting Machine

ANN Artificial Neural Network

ARIMA Auto-Regressive Integrated Moving Average

LSTM Long Short Term Memory

GRU Gated Recurrent Unit

SVR Support Vector Regression

SVM Support Vector Machine

KNN K-Nearest Neighbors

THI Temperature-Humidity Index

WCT Wind-Chill-Temperature

SI Seasonal Index

FI Feature Importance

1 Introduction

In recent years, the European electricity market has been subject to fundamental changes, partly due to geopolitical instability. Currently, electricity prices are highly volatile and affected by geopolitical developments, weather circumstances, and supply disruptions, causing intensified pressure on households and the electricity market as a whole [1]. This has, in turn, lead to an increased need for customers to understand and manage their electricity consumption and future electricity costs more efficiently.

The emergence of environmental problems due to climate change is another reason for sustainable energy management becoming increasingly important [2, 3]. The ability for customers to understand and influence their household energy consumption is significant for sustainable energy use. Household energy consumption makes up a considerable part of the total energy consumption, approximately 30% in some European and American countries. Managing household energy more efficiently has large potential in saving energy, since it is estimated that 27% of energy used by households can be saved through more efficient use [4]. Along with the increased need of efficient energy management, the total demand of electricity in the world is also expanding, furthering the need of electricity load forecasting to support decision-making regarding energy [5, 6]. The above reasons contribute to making the topic of forecasting electricity consumption important to research.

This thesis specifically explores monthly electricity load forecasts for individual residential households. Monthly forecasts are a valuable tool for managing smart-grids and making informed decisions in energy planning and usage [7, 8]. Individual customers can benefit from accurate forecasts by consuming electricity more efficiently, both saving money and resources. Performing these types of forecasts is an intricate task, both due to uncertainty in human behavior, and due to factors like economy, weather, and seasonality affecting the monthly energy consumption [8, 9]. For these reasons, forecasting should be performed using robust and reliable techniques.

Previous research has identified various models for predicting electricity consumption, both statistical and Artificial Intelligence (AI) based. In recent years, increasing interest has been shown for AI solutions to energy consumption forecasting, due to their strength in handling nonlinear problems [10, 11]. Forecasts are often based on several factors such as weather conditions and building conditions, causing nonlinear patterns that statistical methods struggle with, but AI methods handle better [10]. However, one disadvantage with AI methods is the black-box nature of the models which causes a lack of transparency and interpretability in the decision-making process. To solve the problem of elucidating the internal process and decision-making of AI algorithms, the field of Explainable Artificial Intelligence (XAI) has emerged within recent years. XAI methods are able to open up the black box, and provide explanations for why a model predicts a certain way [12].

For users, the ability to have confidence in the forecast is important. Users may, for example, question how the forecasting mechanism makes decisions, or question what factor is the most determining in the forecast. A system that can explain such things to the user is able to increase its reliability with users. XAI techniques can be applied to provide transparency and meaningful insights into the decision-making of complex mechanisms, enabling stakeholders to use AI systems with confidence and accountability [13].

This thesis addresses the problem of monthly electricity consumption forecasting for residential households from two perspectives. Firstly, efforts are made to develop a robust and accurate system for forecasting the electricity consumption of individual households. This system is based on historical electricity consumption data and a diverse set of external variables, with the aim of providing the forecasting models with a wide set of features to base the forecasts on. The forecasts are performed using decision tree ensemble models, known for their robustness and precision in regression tasks [14]. A first research question is posed:

RQ1 – Which decision tree ensemble learning models are the most efficient for forecasting monthly residential electricity consumption?

Secondly, this thesis adopts an explainable AI perspective to enhance forecast interpretability and assess the contributions of both internal (i.e., historical consumption) and external forecasting variables. The SHAP methodology [15], along with traditional feature importance metrics, is utilized to produce insights into how the selected features affect the decision-making of ensemble models. The explainable AI analysis addresses the increased need for transparency and interpretability in AI technologies due their black-box nature, and supports stakeholders in their energy management and forecast development. Previous research has applied XAI approaches for daily and annual energy load forecasting, but this thesis undertakes the novel task of explaining forecasts with a monthly scope. Further, this thesis explores the energy consumption of individual residential households, which has received less attention in previous works in favor of industrial sector forecasts or aggregated forecasts [16, 17]. A second research question is posed:

RQ2 – Which internal and external variables are the most important when forecasting the monthly electricity consumption of residential households using decision tree ensembles, and which variables do not benefit the forecasts?

The experiments conducted in this thesis are based on a real-world dataset provided by an electricity distribution company, Umeå Energi. The dataset consists of historical electricity consumption records collected from residential customers in Umeå, Sweden, aggregated at the monthly level. In addition to consumption measurements, the dataset includes contextual information that enables the integration of external explanatory variables in the form of household attributes. Furthermore, weather data from SMHI[1] and holiday information from Dagsmart API[2] is utilized to provide a wide range of external variables. The use of real operational data allows the proposed forecasting and explainability framework to be evaluated under realistic conditions that reflect the complexities of energy demand in practice. For Umeå Energi, this thesis will add value by suggesting a framework that supports their customers in managing their energy consumption more efficiently. According to internal customer analytics, one of the most highly requested features among customers is to receive an estimation of both the electricity consumption and the electricity cost before the invoice arrives.

[1] Swedish Meteorological and Hydrological Institute, SMHI. https://www.smhi.se/data (Accessed 2026-03-17)

[2] Swedish calendar data API, Dagsmart API. https://dagsmart.se/api/ (Accessed 2026-03-16)

The main contributions of this thesis can be summarized as follows:

- Applies a wide range of input features for forecasting electricity consumption using decision tree ensembles, including historical consumption, household information, weather variables, and holiday information.
- Enhances forecast interpretability through means of explainable AI, which is novel for the monthly scope of electricity consumption forecasts. The application of explainable AI analysis addresses growing demands of transparency and trustworthiness in AI models.
- Bridges a gap on research in the residential scope of electricity consumption forecasting by performing forecasts for individual households. Previous research is mainly focused on large-scale aggregated forecasts and industrial forecasts.

2 Background

This chapter presents theoretical background for concepts and methodologies used in this thesis, including an introduction to time series forecasting, ensemble learning models, and explainable artificial intelligence. The chapter also presents previous works related to this thesis, with a focus on works utilizing ensemble learning techniques and explainable artificial intelligence for energy consumption forecasting.

2.1 Time Series Forecasting

The problem of forecasting electricity consumption can be categorized as a time series forecasting problem. A time series is a set of chronologically arranged observations, generally under the assumption that time is a discrete variable [18]. In a time series with T real value samples $x_1,\ldots,x_T$, the value at time i is represented by x_i ($1 \leq i \leq T$). The problem of time series forecasting can be defined as predicting the values of $x_{w+1},\ldots,x_{w+h}$, where w is the historical window, and h is the prediction horizon. The historical window corresponds to the number of past observations used to generate the forecast, and the prediction horizon defines the number of future time steps to be predicted. Prediction happens given the previous samples $x_1,\ldots,x_w$ ($w+h \leq T$), with the goal of minimizing the error between the predicted value $\hat{x}_{w+i}$ and the actual value x_{w+i} ($1 \leq i \leq h$) [19].

A time series may be univariate or multivariate, where univariate refers to a series with a single observation recorded over time, and multivariate refers to a series with a group of variables that may exhibit interactions. Analysis of time series often begins with obtaining the underlying patterns of the observed data. The next step is fitting a model to represent the data for future predictions, which can be a complex task. The analysis of the time series is usually performed by decomposing it into three components [11, 20]:

1. *Trend* - The general movement of the variable during the observation period. Trend does not take irregularities and seasonality into account.
2. *Seasonality* - The periodic fluctuation of the variable. It includes stable effects, along with time, magnitude, and direction.
3. *Residual* - The remaining variation, a largely unexplainable part of the time series. In some instances, this can be high enough to mask the trend and seasonality.

Both linear and nonlinear techniques may be used for solving time series problems. In linear methods, the idea is that strong correlations in the data allow for linear combinations to determine the next observation. However, the random component of the time series may prevent precise predictions. Nonlinear methods are applied in the machine learning domain to forecast future values based on a model describing the data. Machine learning solutions to forecasting problems have gained popularity, due to the fact that they are suitable both for linear and nonlinear problems [19]. Time series analysis has been extensively applied to the problem of energy consumption forecasting, especially since real-time monitoring of buildings has become more common, and the availability of recorded data has increased [11].

Based on the forecasting horizon, energy consumption forecasting problems are generally divided into three categories: short-term forecasting (1 hour to 1 week), medium-term forecasting (1 month to 1 year), and long-term forecasting (1 year and above) [10]. This thesis explores medium-term forecasting with a horizon of 1 month.

2.2 Ensemble Learning for Energy Forecasting

Research in the field of electricity consumption forecasting has established many methods for producing forecasts. These methods can roughly be divided into two categories; conventional methods and AI-based methods. Conventional methods such as time series analysis and regression methods have previously been widely applied to solve the problem of electricity consumption forecasting. Nowadays, AI-based methods are more popular due to their strength in solving nonlinear problems, something that the conventional methods struggle with [10]. Overall, two of the most widely applied models in the field are artificial neural networks (ANN) and support vector machines (SVM), along with the statistical autoregressive integrated moving average (ARIMA) model [17]. This thesis utilizes less widely used, although robust and precise, ensemble learning techniques to perform forecasts. This section introduces theoretical background as well as previous works on ensemble learning for energy consumption forecasts.

Ensemble learning is a family of methods that combine several Machine Learning (ML) algorithms to make decisions. It can be seen as the ML interpretation of “wisdom of the crowd”, that is, the idea that aggregating the opinions of several individuals is better than selecting the opinion of one individual. Multiple weak learners make predictive results that are fused together via voting mechanisms, in order to achieve better performance than from any single algorithm. Any type of machine learning algorithm, e.g. decision trees, neural networks, or regression models, can be used as ensemble learners [21, 22]. Ensemble methods can be categorized into parallel and sequential methods, where parallel methods train base classifiers separately and combine the individual predictions via a combiner, and sequential methods train models iteratively to correct the errors of previous models. The bagging method and its extension, Random Forest (RF), is a popular parallel algorithm, while the boosting algorithm is a popular sequential method [23]. This thesis utilizes both approaches. Figure 1 shows block diagrams for parallel and sequential ensemble learning.

According to Dietterich [24], there are three reasons why ensemble learning methods achieve good performance in machine learning. The first reason is statistical. Models can be seen as searching for the best hypothesis within a hypothesis space H. When data is limited, which it often is, the models can find several hypotheses in H that give the same accuracy on the training data. This makes it hard to choose between hypotheses, since it is unknown which one will generalize better for unseen data. Ensemble learning can help with avoiding this problem of overfitting. Secondly, ensemble learning provides computational benefits. Many models function via performing some type of local search, which may

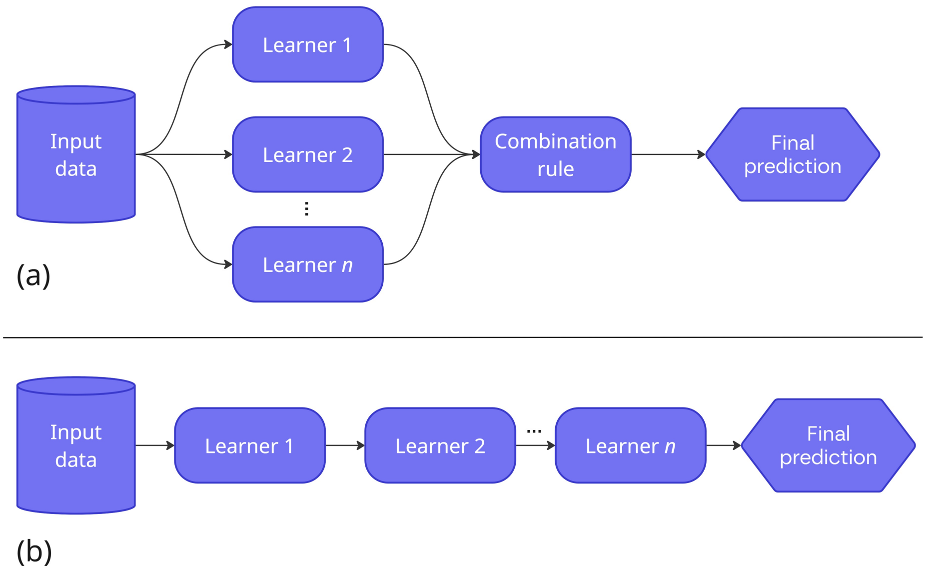

Figure 1: Block diagram showing (a) parallel ensemble learning and (b) sequential ensemble learning [23].

be prone to getting stuck in a local optima. A better approximation of the unknown true function may be obtained from an ensemble constructed by the local search started from many different points. The third reason is representational. In most situations, the unknown true function may not be included in H. Combining several hypotheses from H, the space of representable functions can be expanded to possibly include the true function.

Three common strategies for constructing ensembles are *bagging*, *boosting*, and *stacking* [14]. Bagging, or bootstrap aggregating, proposed by Breiman [25] is a method for obtaining an aggregated prediction by building multiple versions of a predictor. Bootstrapping is used to generate multiple samples of data from the original dataset, which are then used to independently train the different predictors. Majority voting is used to aggregate the final prediction [21]. The main objective of the bagging technique is to reduce the variance of a prediction model. One common example of the bagging technique is the random forest algorithm [14, 26]. Equation 2.1 formally defines a bagging regressor, where $f_m(x)$ are induced regressors [27].

$$F(x) = \frac{1}{M} \sum_{m=1}^{M} f_m(x) \tag{2.1}$$

In boosting, several models with individually weak predictions are aggregated to obtain a model with high accuracy. Models are trained sequentially to correct the errors of previous models. The main objective with this technique is reducing model bias [14, 26, 28]. Two examples of boosting algorithms are Gradient Boosting Machines (GBM), which uses gradient descent to minimize the loss function, and XGBoost, which is an extension of GBM focused on resource optimization and prevention of overfitting [26]. Equation 2.2 formally defines a boosting regressor, where f_m represents the induced regressors and α_m represents their respective weights [27].

$$F(x) = \sum_{m=1}^{M} \alpha_m \cdot f_m(x) \tag{2.2}$$

Stacking utilizes different learning algorithms called base learners to make initial predictions. These outputs are then combined via a meta-learner to make the ultimate predictions. The goal is to capture a wider range of patterns using the strength of different models, while minimizing the generalization errors [14, 19, 26]. Accuracy improvements mainly happen when there is diversity among the individual models, since differences in generalization principles generally lead to different results [26].

2.2.1 Decision Tree Ensembles

This thesis specifically focuses on decision tree ensembles to produce forecasts. Decision trees classify data items by asking a sequence of questions about the items' features. The nodes of the tree contain questions, and for each possible answer, every internal node points to one child node. The sorting of an item into a class begins from the root of the tree, and according to the item's answer to the questions, it follows the path to a leaf node [29]. An example of a decision tree structure is found in Figure 2. One large advantage with decision trees is that they are considerably easier to interpret than neural networks, as they follow logical rules instead of having connections with numeric weights between nodes. Decision trees perform well for a few highly relevant attributes, but not as well if there are many attributes with complex interactions.

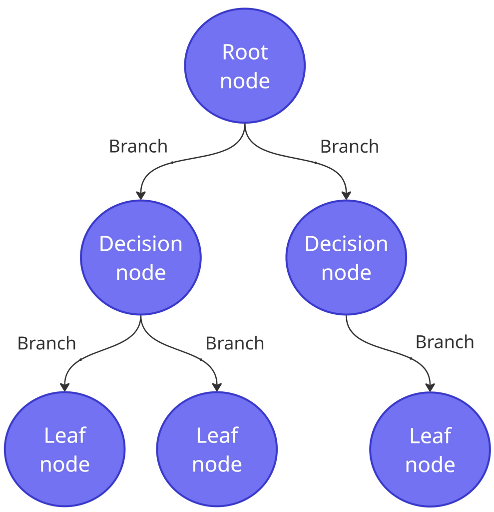

Figure 2: A decision tree.

Ensembles of decision trees can be employed to increase accuracy when the training data is more complex and changing [30]. Decision tree ensembles are considered among the most reliable and precise methods for regression and classification tasks, and they have been shown to be effective for predicting energy consumption [14]. However, one disadvantage is that the interpretability decreases when multiple classifiers are involved, making it more difficult to understand the reasoning behind the ensemble's decision [30]. This creates a need for model explainability outside the ensemble's inner workings, which in this thesis is addressed by applying explainable AI analysis.

2.2.2 Previous Works

Ensemble learning has previously been applied for the task of electricity consumption forecasting, mostly in the short-term scope. Divina et al. [19] proposed a stacking ensemble learning approach for short-term forecasting of Spain's general electricity consumption. They used a two-layer approach based on a bottom and a top layer. The bottom layer consisted of three base learning methods; regression trees based on evolutionary algorithms, Artificial Neural Network (ANN), and Random Forest (RF). The top layer used GBM to combine the predictions of the base learners, and produce the final predictions. They found that the combination of weaker learning methods was able to produce superior results compared to several baselines, showing that ensemble learning is suitable for short-term electricity consumption forecasting. Neural networks were able to predict with high accuracy in the very near future, but accuracy declined when predicting further in the future. The ensemble method also showed significantly better performance compared to baselines when the amount of historical data was small, indicating that ensemble methods may be more flexible for different quantities of data.

Pinto et al. [31] also utilized ensemble learning for forecasting electricity consumption in the short-term. Using consumption data from an office building, along with other variables such as temperature and humidity, they predicted hour-ahead consumption with three different ensemble methods. These methods were random forests, gradient boosted regression trees (GBR), and AdaBoost. They found that ensemble learning methods, especially AdaBoost, outperformed benchmark models such as support vector regression and fuzzy rule-based methods. They highlight that including temperature and humidity information is advantageous to the forecasts, and they suggest that future work should consider additional external variables such as solar irradiation and thermal sensation.

Khan et al. [32] presented a stacking deep learning ensemble for short-term forecasts. The lowest level of the ensemble was compromised of multiple individual Long Short Term Memory (LSTM) models responsible for producing outputs. The top level of the ensemble was a Gated Recurrent Unit (GRU) responsible for increasing accuracy by combining outputs from the previous layer. They also utilized clustering to identify customer groups with similar profiles and load patterns. Based on hourly residential consumption data, they predicted hourly, daily, and weekly consumption, and found that the proposed model outperformed benchmark deep learning models. The clustering of customers helped with reducing model errors. Additionally, it was found that their model performed best for hour-ahead forecasts, while errors increased when approaching the weekly scales.

In another short-term study by Touzani et al. [33], the energy consumption of commercial buildings was predicted using the decision tree-based GBM model. They emphasized the flexibility and robustness of decision tree ensembles, particularly GBM, when the number of input parameters is high. Results showed that GBM outperformed a linear regression model and a random forest model in 80% of cases.

Hadjout et al. [34] used an ensemble deep learning approach to forecast monthly electricity consumption for high voltage customers in the Algerian economic sector. They utilized three different deep learning models – LSTM, GRU, and Temporal Convolutional Networks (TCN) – and applied grid search to find optimal weights for each model in the final ensemble prediction. Compared to the individual models separately, the ensemble approach performed significantly better according to Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) metrics.

Long-term forecasting using ensemble learning has been researched by Chen et al. [35]. They predicted the annual electricity consumption for American households using extreme gradient boosting forests and deep networks as base learners, and ridge regression as a combiner method. The data included variables pertaining to household demographics, household type, appliances, and geographical data, providing a wide range of information. The ensemble framework was able to reduce errors by 30% compared to classical regression models such as linear regression and support vector machines. Investigation into the importance of input features using principal component analysis showed that electricity price, number of rooms, natural gas usage, and temperature during summer were among the variables that contributed the most information. They also note that some features, like number of rooms, number of bedrooms, and total square footage, have a strong correlation and have potential to be represented more efficiently. They argue that feature selection is necessary, since discarding of unimportant features can enhance model performance.

These works demonstrate the strength of ensemble models in forecasting electricity consumption, particularly when data is diverse and changing. Consequently, the choice of ensemble models in this thesis is supported, since the electricity consumption data is sourced from many households with different characteristics in combination with external data.

2.3 Explainable AI Techniques

Apart from utilizing robust ensemble models to forecast electricity consumption, this thesis applies explainable AI techniques to increase model interpretability. In recent years, artificial intelligence has been widely adopted in society, and humans increasingly rely on AI as part of decision-making. Research in the field of AI has for many years focused on improving the predictive accuracy. However, the black-box nature of many AI systems creates a lack of transparency, and explaining to users how the AI systems work is highly problematic. Pressure from society, ethics, and legislation has created a demand of a new type of AI that is interpretable to users and can explain its inner functions [12, 36].

Machine learning models have two main objectives; firstly, they should be able to make accurate predictions for new given input features, and secondly, they should be interpretable (i.e., provide explanations for how a given input relates to the output). Usually, models have a trade-off between accuracy and interpretability. Linear regression models are generally transparent and easy to understand, but their performance is often worse than more complex nonlinear models. On the other hand, the nonlinear models perform well but lack in interpretability [37].

Explainable AI (XAI) methods that elucidate the decision-making process have been developed to enhance the interpretability of AI models. These methods can be applied at different levels of the AI modelling. Firstly, explainability can be applied pre-modelling through means of data pre-processing and data analysis. This approach gives insights into the datasets used for training the ML models, and it can expose imbalances in the data. Secondly, the model itself can be inherently interpretable. This refers to models that have a simple structure, such as linear regression and k-Nearest Neighbors. Models like these are understandable by design, and users are able to interpret the model based on the model summary or parameters. Thirdly, explainability can be applied post-modelling to deal with the black-box problem of more complex ML models. Post-modelling methods include textual and visual justifications, simplification of the model, and feature importance, as illustrated in Figure 3. These methods can explain how an algorithm performs during training, and how predictions are generated given a certain input [12, 36].

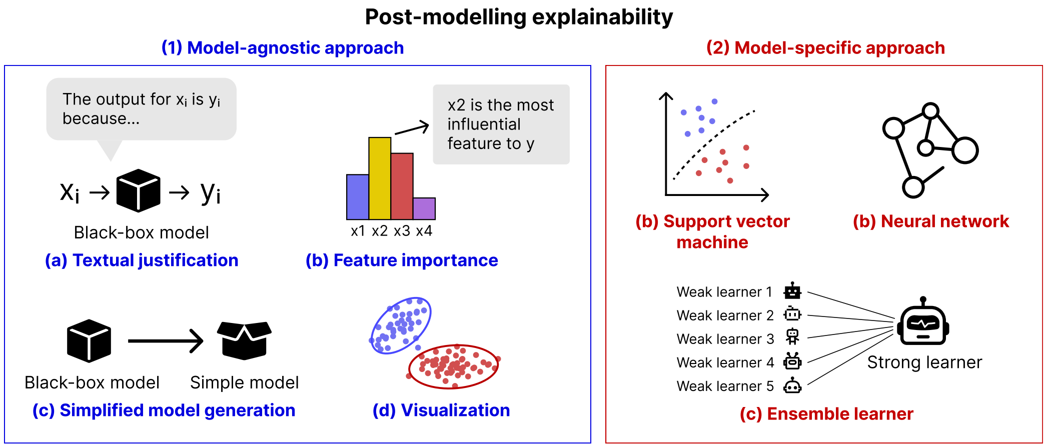

Figure 3: Two categories of post-modelling explainability, including (1) model-agnostic approaches and (2) model-specific approaches [12].

XAI approaches may be *model-specific*, meaning they are applied to a specific type of model, or *model-agnostic*, meaning they are not dependent on model architecture and can be applied for any model. Additionally, XAI explanations can be *local*, which means that individual predictions from given feature values are explained, while *global* explanations mean that feature contributions are explained for all instances [36, 38].

Feature importance is the most common explanation tool for classification models [37]. It describes how important a variable was for the model output, that is, the individual contribution of a feature, regardless of the shape or direction of the feature's effect, dependent on a particular classifier. Feature importance reveals which features are crucial for the model when making predictions. The feature importance descriptions may be local or global, where local importance describes the feature's contribution to a specific prediction, and global importance describes the feature's contribution for the entire model [12, 36, 37].

Ensemble models, used in this thesis, are based on the aggregation of several models, which makes it more complex to interpret the results. Since the models are not interpretable by design, there is a need for the implementation of post-modeling XAI methods to understand the models' predictions. Among the post-modeling XAI methods, simplification and feature relevance are the most widely used for this task [12].

2.3.1 SHAP

SHapley Additive exPlanations (SHAP) is a popular method for model explainability, and it is the selected XAI technique for this thesis. The technique is model-agnostic and can therefore be used for any ML model. The SHAP method has its foundation in game theory, where the contribution of each player to the total payoff is calculated. In the case of machine learning, the players are represented by features (variables) and the payoff is the model output. To calculate the contribution score for each feature, all different combinations of features (i.e., coalitions in game theory) are evaluated. The Shapley value of feature *i* can be calculated according to Equation 2.3, where ϕ_i is the relative contribution to the model output, M is the set of all features, and $f_S(x)$ is the model output for feature subset S [38, 39].

$$\phi_i = \sum_{S \subseteq M \setminus \{i\}} \frac{|S|!(|M|-|S|-1)!}{|M|!} [f_{S \cup \{i\}}(x) - f_S(x)] \tag{2.3}$$

A positive Shapley value for a feature means that the feature contributes to increasing the predicted value. For example, when predicting electricity consumption, a substantially positive Shapley value for temperature means that the increasing temperature contributes to an increased electricity usage. SHAP can be used to explain model outputs both locally and globally, i.e., both for specific instances and for all instances [14, 38].

When using SHAP for model explanations, it is important that users are aware of some important considerations of the method. Firstly, the SHAP method is model-dependent, meaning it can produce different explanations for the same task when different models are applied. Secondly, the explainability scores assigned to features do not represent the weights of the features. Rather, the explainability lies in the ranking of the features, meaning that feature importance is deduced from the ordering. Thirdly, it is assumed that the features are independent of each other. Therefore, some features that are correlated with other features may receive low scores even though they contribute significantly to the output. This is because the correlated features have already accounted for the impact of the low-scoring feature. Lastly, if SHAP is applied to a biased classifier (e.g. due to data bias or algorithmic bias), the method may produce unrealistic explanations that do not describe the bias. Because of these points, it is crucial to present SHAP results in an informative way by including corresponding output plots and assumptions behind the method [38].

2.3.2 Previous Works

Explainable AI methods have previously been applied for the task of short-term electricity consumption forecasting. Sim et al. [3] proposed a novel XAI approach for variable selection in energy consumption forecasting. They analyzed the influence of 15 different input variables pertaining to time information, climate data, and past energy consumption data. LSTM was selected as prediction model and SHAP was used to interpret the variables. The findings showed that time data such as year and hour had a strong influence on the prediction results, along with temperature, surface temperature, and energy-consumption-difference. Variables with a weak or ambiguous influence included humidity, wind speed, and sunshine amount. They concluded that forecasting performance can be significantly improved by selecting high-impact variables based on XAI analysis.

Maarif et al. [6] investigated the effect of 9 different features on energy consumption forecasting for a South Korean steel company. The features provided information about historical consumption, lagging and leading current power, carbon dioxide emissions, and load type. They utilized various LSTM models to make hour-ahead forecasts, and they used SHAP for the XAI analysis. It was found that leading current reactive power and seconds from midnight strongly influenced the model output, while CO_2 and load type had no effect on the forecast. They note that it is important to consider interpretability and performance trade-offs when developing XAI forecasting models, and that further research into this topic is needed for understanding the applicability of these models in the energy sector.

Another hourly electricity consumption forecasting study was performed by Lazzari et al. [16]. They provided a novel perspective by incorporating user behavior to enhance the accuracy of residential energy consumption forecasting. User behavior clusters were identified, and the day-ahead behavior was predicted using an XGBoost model. The predicted behavior, along with historical consumption, was then provided to an ANN model responsible for predicting the hourly day-ahead consumption. Compared to an ANN without the user behavior input, the proposed model was able to reduce MAPE and NRMSE by 7% and 9%, respectively. A feature importance analysis using SHAP showed that calendar variables, namely week and season, had the greatest importance for the model. Temperature, notably minimum and maximum daily temperature, also had a large influence on model output.

Moon et al. [14] utilized ensemble learning and XAI methods for short-term residential building electricity consumption forecasting. They performed an extensive comparison between five ensemble learning models and ten deep learning models. It was found that the decision tree-based ensemble methods were superior in handling diverse and noisy data, rendering them useful for electricity consumption forecasting. They highlight robustness, interpretability, and efficiency as advantageous characteristics in ensemble models. They argue that ensemble models can handle data that is not extensively preprocessed and that they require less data than deep learning models. A feature importance and SHAP analysis was performed on several ensemble models, revealing that Temperature-Humidity Index (THI) and Wind-Chill-Temperature (WCT) (see section 3.1.2 for definition of THI and WCT) had significant importance for the forecasting. The SHAP analysis combined with the decision tree-based models provided interpretive insights about complex interactions in the data.

XAI approaches have also been applied to long-term forecasting [40]. Chakraborty et al. [41] predicted long-term cooling energy consumption for residential and commercial buildings under different climate change scenarios. They used SHAP to explain the results and concluded that the XAI method increased the interpretability of the XGBoost prediction model. Another long-term study was conducted by Wenninger et al. [42], who predicted the annual energy consumption for German households, and proposed a novel QLattice model for enhanced explainability. They found that the QLattice model was appropriate for the task of energy consumption forecasting, even though XGBoost achieved slightly better predictive performance. However, a significant strength with the QLattice algorithm is its transparent design, which was found to greatly increase the model explainability compared to established models such as ANN, XGBoost, and Support Vector Regression (SVR).

Even though XAI techniques have been applied in the field of electricity consumption forecasting, there is a clear gap in the research, especially pertaining to the medium-term horizon of forecasts. This thesis addresses the gap by utilizing the SHAP methodology to interpret the forecast output from five different decision tree ensemble models, using a wide range of input features in the monthly scope.

2.4 Monthly Forecasting

A perspective that makes this thesis stand out from many previous works in the field is the focus on monthly forecasts, a perspective that has previously received less attention than short- and long-term forecasts. Forecasting electricity consumption in the monthly scope requires handling complex non-linear patterns and seasonal tendencies [43]. Monthly consumption is affected by factors such as economy, temperature, and seasonality, which makes the consumption exhibit a mixture of trend, periodicity, and randomness [7]. Additionally, when predicting the consumption of small-scale users, like in this thesis, the power demand is more sensitive to random factors. Human behavior may change periodically, both due to social factors such as holidays and summer vacations, and due to natural factors like the change in seasons [9].

2.4.1 Previous Works

To address the seasonality of monthly forecasting, models incorporating seasonal indexing (SI) have been proposed. Cao and Wu [43] aimed to improve monthly electricity consumption forecasts using support vector regression (SVR) in combination with SI and a novel evolutionary algorithm called fruit fly optimization (FOA). They calculated the seasonal index for each month, which was then used to adjust the predicted values according to the seasonal tendencies. FOA was applied to find the best values for parameters of the SVR model. Their results showed that both SI and FOA was able to effectively improve the forecasting accuracy, suggesting that the hybrid approach is a viable option for monthly forecasting.

Tang et al. [7] proposed a model specifically designed to handle sparse data with periodic properties. Their model also incorporated SI to address the seasonal component. A GM(1,1) gray model, a mathematical optimization model for incomplete data, was used to predict the seasonal index. The SI was combined with a trend forecast value to obtain the final prediction, and results showed that the model significantly improved accuracy, even for small electricity consumption datasets.

Sun et al. [9] approached the monthly forecasting problem by seasonal and trend decomposition of the electricity load time series, in combination with Auto-regressive Integrated Moving Average (ARIMA). This technique allowed for control of both the seasonal component's change rate and the trend component's smoothness, as well as increased robustness to outliers. Additionally, they incorporated economic factors into the forecast to improve accuracy. They found that the proposed model was able to increase forecasting accuracy, and determined that both the decomposition approach and inclusion of economic variables were effective to the forecast. For future improvements, they suggested developing detailed consumption categories and human behavior models to better account for human factors and possibly improve accuracy.

2.5 Residential Electricity Consumption Forecasting

Another factor that makes this thesis stand out is that it forecasts the electricity consumption for individual residential households. Forecasting for residential buildings is less explored, since most previous works focus on forecasting for the industrial sector, or focus on forecasting the aggregated consumption on a large scale. Although underexplored, household energy forecasts, in particular short- and medium-term forecasts, are important for managing and planning operations and future grids. They are also important for the energy management of individual households, since there exists potential for households to save substantial amounts of energy through better management [4, 8, 17]. In contrast to the large-scale forecasts, forecasting for a single household is challenging due to factors of uncertainty. The energy load on a larger scale follows patterns, but the energy consumption of an individual household is often inconsistent [8].

2.5.1 Previous Works

Kong et al. [8] forecasted the short-term electricity consumption for 69 Australian households. They selected LSTM as their forecasting model, and applied density based clustering to evaluate inconsistencies in the households' consumption patterns. Findings revealed that the proposed LSTM framework was able to outperform multiple benchmarks such as K-Nearest Neighbors (KNN) and Back-propagation Neural Network (BPNN). Additionally, when forecasting the aggregated energy load, they found that forecasting individual loads and then aggregating them produced better results than directly forecasting the aggregated load. They concluded that results could be improved via parameter tuning, especially for households with high volatility in their electricity load.

Kiprijanovska et al. [44] also forecasted individual household energy consumption in the short-term scope. They proposed a deep residual neural network with external features and historical consumption. Special domain-specific time-series features were also included to better model the household consumption. The proposed model showed better performance than end-to-end deep learning models, largely due to the domain-specific features. They suggest future work should employ clustering of households to create consumption profiles, due to the large variations in consumption at the household level. This thesis tries to address this problem by performing K-means clustering on the electricity consumption.

3 Methodology

[Should change entire method to past tense]

This chapter describes the experimental methodology applied to answer the research questions. The experiments aimed to develop monthly electricity consumption forecasts for residential households, with a focus on forecast explainability. A variety of input variables, including historical electricity consumption, household information, weather variables, and calendar information, were used to provide diverse information to the forecasting models. Five decision tree-based ensemble learning models were utilized for the forecasting, and their predictive performance and explainability were assessed and compared. The five models were chosen for their reliability and robustness in forecasting, and for their various unique strengths [14]. The explainable AI method SHAP was used to investigate the effect and importance of the model features. Figure 4 shows the general architecture of building the explainable electricity consumption forecasts.

3.1 Data Preparation and Analysis

This study utilized electricity consumption and household data from Umeå Energi, along with external variables in the form of weather data and holiday information. This section described the data, presented analytical insights as part of enhancing the explainability of the forecasts, and detailed how the data was preprocessed before performing forecasts.

3.1.1 Electricity Consumption Data

Swedish electricity company Umeå Energi supported this research with a tailored dataset containing monthly electricity consumption from 14 286 Swedish households. The data consisted of 4 years of measurements retrieved from January 1st 2022 to November 30th 2025. For the purpose of this research, households were defined as unique customer and facility combinations (i.e., one customer may have had several facilities, which were regarded as separate households). Households that had any undefined or less than 10 kWh consumption measurements during the time period were removed from the dataset, leaving 13 692 households with complete measurements. All customers were private customers located in Umeå in the northern part of Sweden, meaning the data represented household electricity consumption in a cold climate. The dataset is summarized in Table 1, and Figure 5 shows the distribution of the electricity consumption for all households during the entire time period.

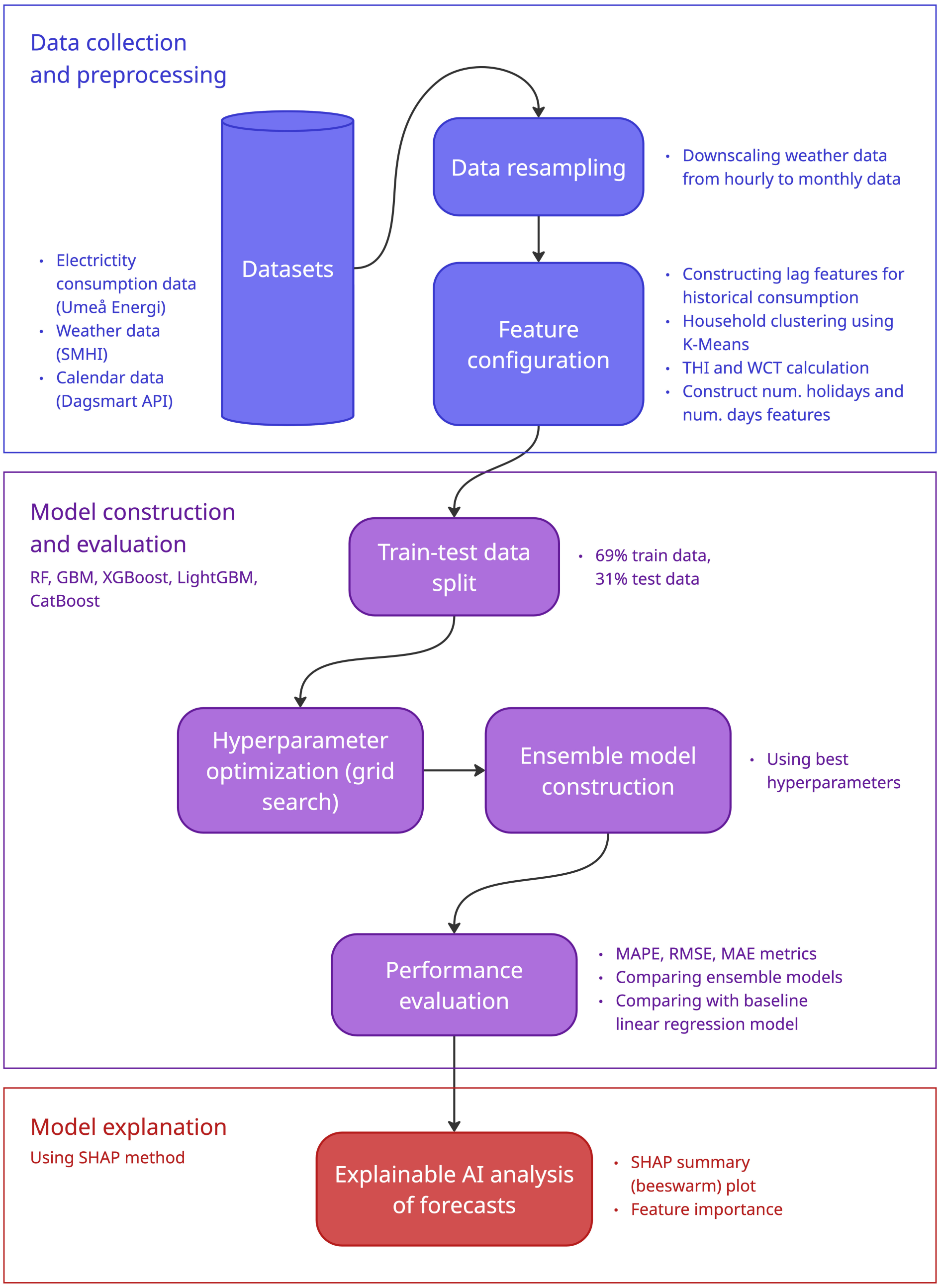

Figure 4: Architecture of modelling explainable electricity consumption forecasts.

Table 1 Summary of the Umeå Energi electricity consumption data.

Statistic	**Umeå Energi Data (kWh)**
Minimum	10
First quartile	124
Median	220
Mean	465
Third quartile	489
Maximum	13 490
Time range	2022-01-01 to 2025-11-30
Location	Umeå, Sweden
Number of households	13 692

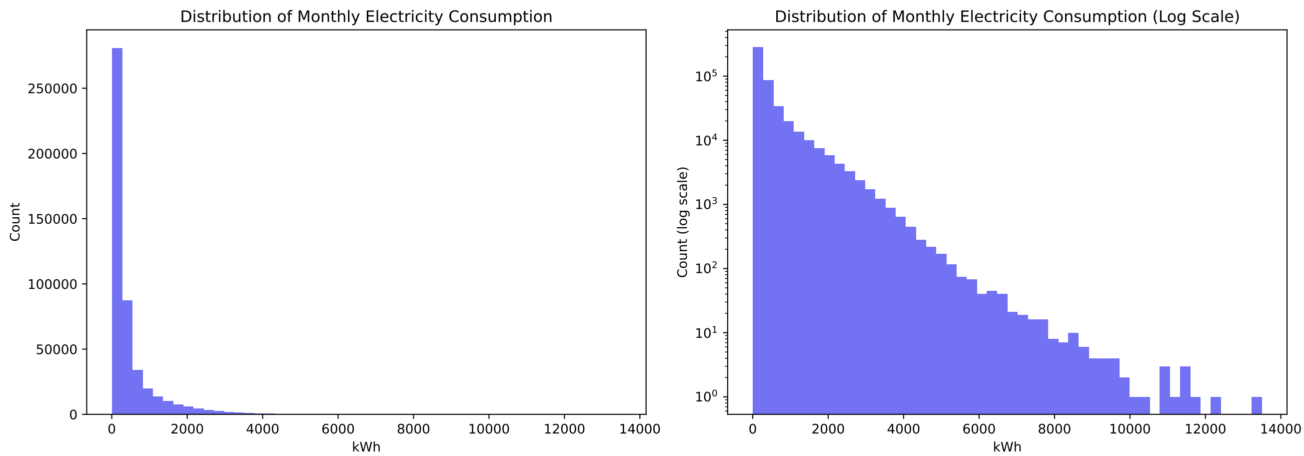

Figure 5: Histogram over the monthly electricity consumption distribution from January 2022 to November 2025.

Table 1 and Figure 5 show that a vast majority of households consume electricity in a similar low range per month, but that there are outliers where the monthly consumption exceeds several thousand kWh. Figure 6 shows the average monthly consumption for all households. It is clear that there is a significant seasonal pattern in the consumption, with more electricity being consumed in the winter and less in the summer. This makes sense for the cold climate, since the winter requires more heating for households with electrical heating, while summers are cool enough to not require much electrical cooling. A warmer climate may have the opposite consumption pattern, since warm summers will have a higher need for cooling measures [41, 45]. Figure 7 illustrates the average consumption by month, showing that consumption varies more during the winter months, while the average consumption is more stable during the summer.

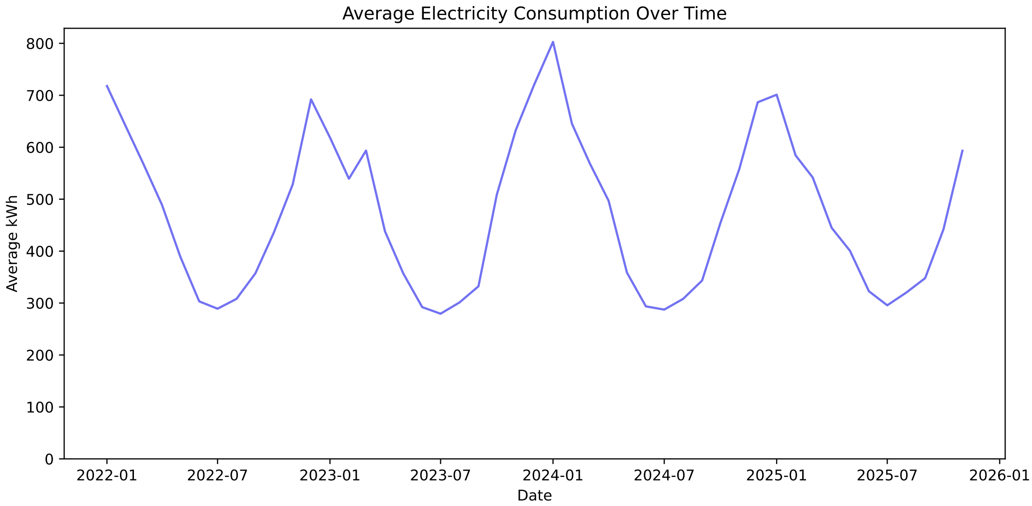

Figure 6: Average household electricity consumption from January 2022 to November 2025.

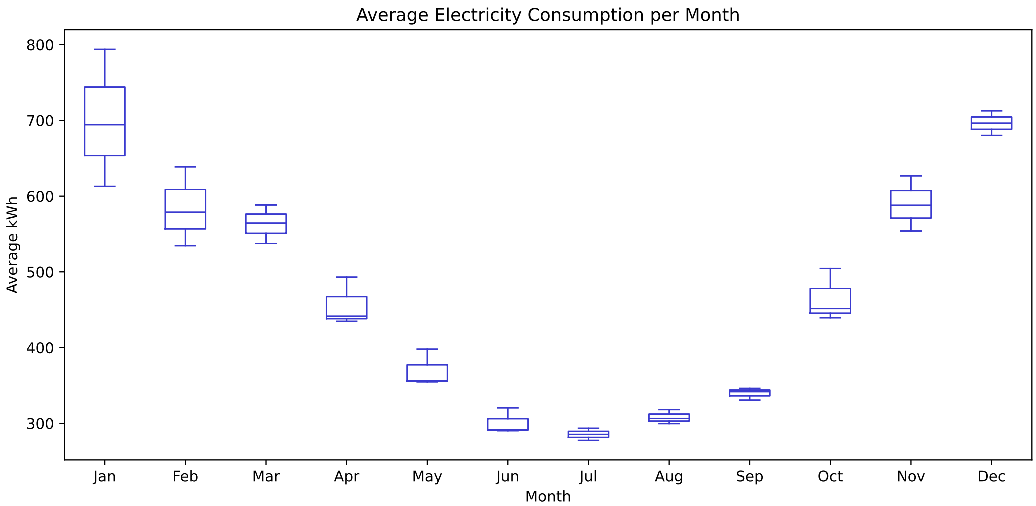

Figure 7: Average household electricity consumption by month from January 2022 to November 2025.

Three input features were constructed from the historical electricity consumption; lag_1 which represented the consumption of the previous month for the target household, lag_12 which represented the consumption for the same month the previous year for the target household, and $rolling\_mean\_3$ which represented the average consumption over the past three months for the target household. These variables express the recent consumption patterns, but also the monthly dynamic. Since the year 2022 did not have any past year historical consumption to refer to, it was removed from the training data.

Household Information

In addition to the historical electricity consumption, the Umeå Energi dataset contained information about the households. Households were labeled by type of residence, type of heating system, and type of contract agreement with Umeå Energi. The type of residence was either house, apartment, or vacation home. Heating could be electrical, non-electrical, or district heating. Contract agreements were either variable, fixed, or default assigned contracts. The share of households in each category is found in Table 2. Any given household could change between categories over time (e.g., change its type of heating system), and therefore, the household information is updated per month. Subsequently, Table 2 is not presented in a number of households per category, but rather, the share of observations that occupy a given category.

These features are not directly maintained by Umeå Energi, but instead, the labeling is done by Umeå Energi based on certain properties of the customer, which may cause the labeling to be inaccurate in some cases. However, it provides a good estimation of the household properties.

Table 2 Household types in the Umeå Energi dataset.

Type	Percentage
Apartment	58%
House	38%
Vacation home	4%
Non-electrical heating	70%
Electrical heating	19%
District heating	11%
Variable contract	82%
Default contract	18%
Fixed contract	$< 1\%$

Household Clustering Using K-Means

Since there were many households in the electricity consumption data, it was relevant to not only group households by their properties, but also by their consumption patterns. The idea was to identify groups where the similarity within the group is higher than that among the groups [32]. K-Means clustering is one of the most popular algorithms for solving this problem in a simple way. K-Means is an unsupervised learning algorithm, where the main idea is to select k centroids, one for each cluster, and place them as far from each other as possible. The next step is to associate each point in the dataset with the closest centroid. After associating all points, the k centroids are recalculated by averaging the points within each cluster. This process is repeated until the centroids no longer move. Formally, the algorithm aims at minimizing the squared error function $W(S,C)$ defined in Equation 3.1 [46].

$$W(S,C) = \sum_{k=1}^{K} \sum_{i \in S_k} \|y_i - c_k\|^2 \tag{3.1}$$

In the equation, S is a partition of k clusters represented by observations in the form of vectors y_i. Each cluster S_k is non-empty and non-overlapping, and has a centroid c_k ($k = 1, 2, \ldots, K$).

In this thesis, K-Means clustering was applied to define household groups based on their electricity consumption. The elbow method was used to find an optimal number of k, i.e. the number of clusters. The elbow method is a visual method, where different values of k are plotted against the variance within the clusters [47]. As the value of k increases, the average variance becomes smaller. The goal is to find a value of k for which the variance stops decreasing sharply, that is, the elbow of the graph. Figure 8 shows the elbow plot for determining the optimal number of household groups, in this case $k = 3$. This resulted in three household groups; low consumption, medium consumption, and high consumption. Each household in the dataset was assigned a label based on the cluster it belongs to, and this label became an input feature for the forecasting models. Table 3 details each cluster regarding size and consumption patterns. The large standard deviation present in each cluster suggests that outlier months with high consumption exist for all identified consumption patterns.

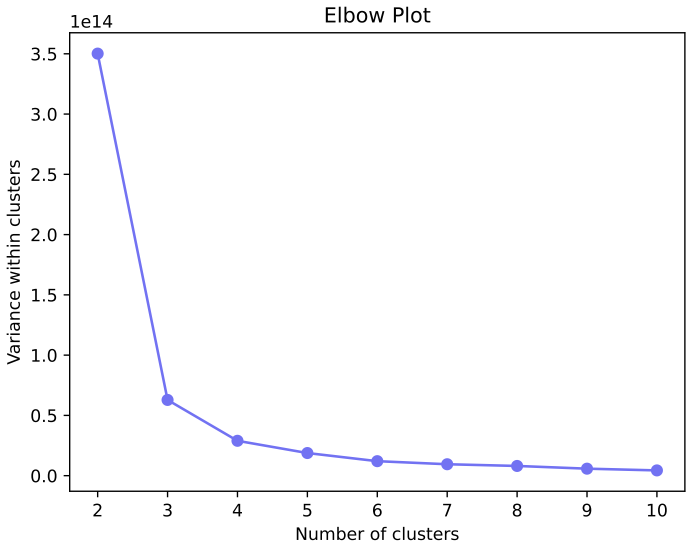

Figure 8: Elbow plot for finding the optimal number of household consumption profile clusters k.

Table 3 Clustered consumption profiles in the Umeå Energi dataset.

Cluster	Households	Mean (kWh)	Median (kWh)	SD (kWh)
Low consumption	1 809	369	174	530
Medium consumption	4 559	419	232	556
High consumption	7 324	528	233	730

Variable Interactions

Interactions between the variables were investigated using a correlation map, as shown in Figure 9. This correlation map includes the internal historical consumption features as well as external features in the form of weather indices (THI, WCT see Equation 3.2 and Equation 3.3) and calendar information (number of holidays, number of days). The three historical consumption variables show a high correlation with each other, while they have a low correlation with all the external variables, indicating that the external variables may contribute information to the forecasts that internal features are not able to capture.

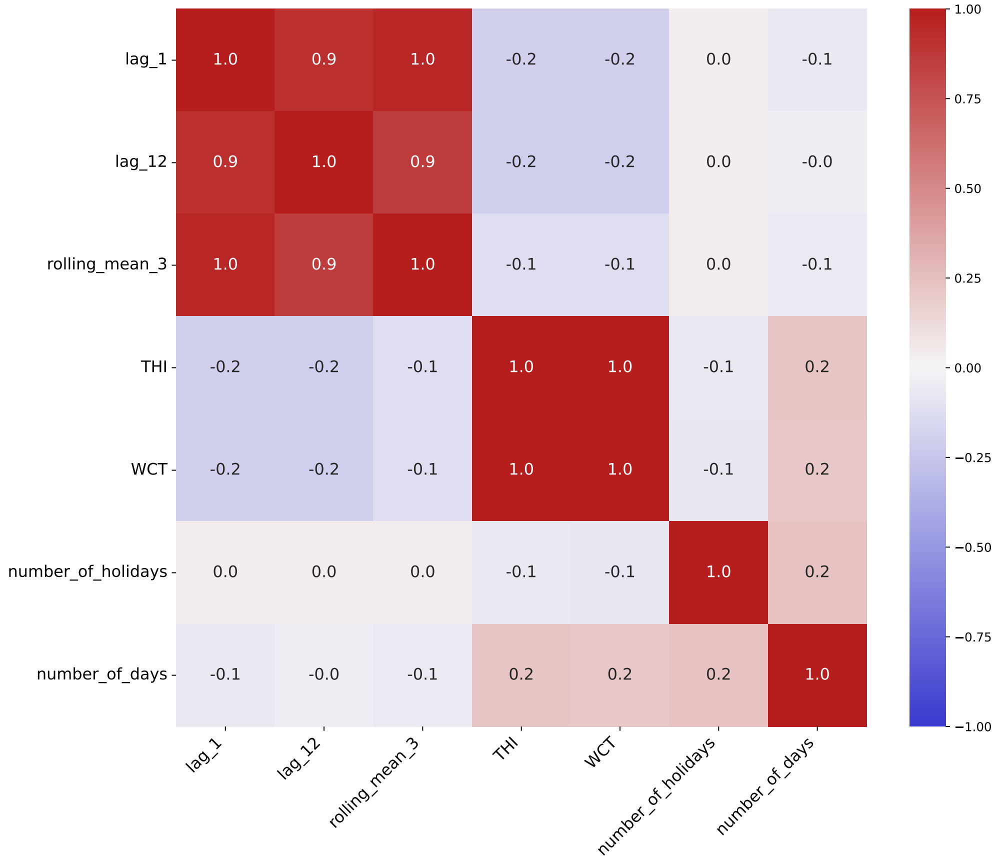

Figure 9: Correlation map of internal variables, weather indices, and holiday indicators.

3.1.2 External Variables

Along with the historical electricity consumption data from Umeå Energi, weather variables were retrieved from SMHI[1] and calendar information was retrieved from Dagsmart API[2]. These external variables were intended to improve the forecasts by including environmental factors that may affect consumption patterns.

[1]Swedish Meteorological and Hydrological Institute, SMHI. https://www.smhi.se/data (Accessed 2026-03-17)

[2]Swedish calendar data API, Dagsmart API. https://dagsmart.se/api/ (Accessed 2026-03-16)

Weather Data

Weather data was retrieved from SMHI's open data for the January 1st 2022 to November 30th 2025 time period. Nine variables were selected from the Umeå Airport measurement station, while three variables pertaining to solar attributes were retrieved from the Umeå Sol station, and snow depth was retrieved from Umeå Röbäcksdalen. All variables were obtained as hourly measurements, except snow depth which was retrieved at daily intervals. Table 4 details the weather variables and their measurement units. Across all variables, 4.9% of the measurements were missing, where most missing values occurred in the lowest cloud base variable that had 42% missing values.

The measurements of all variables were aggregated to monthly values using mean. Figure 10 shows the monthly values of each weather variable plotted over the entire time range. The seasonal component is apparent for many of the weather variables, while some variables such as wind speed and air pressure lack a clear seasonal pattern. This creates potential for the different weather variables to contribute to the forecasts in unique ways.

Table 4 Weather variables retrieved from SMHI.

Variable	Description	Resolution
temperature	Air temperature (°C)	Hourly
dew_point_temp	Dew point temperature (°C)	Hourly
snow_depth	Snow depth (m)	Daily
humidity	Relative humidity (%)	Hourly
wind_direction	Wind direction (°C)	Hourly
wind_speed	Wind speed (m/s)	Hourly
lowest_cloud	Lowest cloud base (m)	Hourly
global_radiation	Global irradiance (Sweden) (W/m^2)	Hourly
longwave_radiation	Long wave irradiance (W/m^2)	Hourly
sunshine	Sunshine amount (s)	Hourly
air_pressure	Air pressure (hPa)	Hourly
visibility	Visibility (m)	Hourly

Although weather factors have been proven to be useful for predicting electricity consumption, the link between weather and electricity consumption can be indistinct. Weather indices incorporating temperature, humidity, and wind speed can provide a better representation of the perceived environmental effects and the impact on electricity consumption [14]. Two indices, temperature humidity index (THI) and wind chill temperature (WCT), were calculated as in Equation 3.2 and Equation 3.3. In the equations, *temp* is temperature, *humi* is humidity, and *ws* is wind speed. These indices were included as input features for the forecasting models.

$$THI = (1.8 \times temp + 32) - [(0.55 - 0.0055 \times humi) \times (1.8 \times temp - 26)] \tag{3.2}$$

$$WCT = 13.12 + 0.6215 \times temp - 11.37 \times ws^{0.16} + 0.3965 \times temp \times ws^{0.16} \tag{3.3}$$

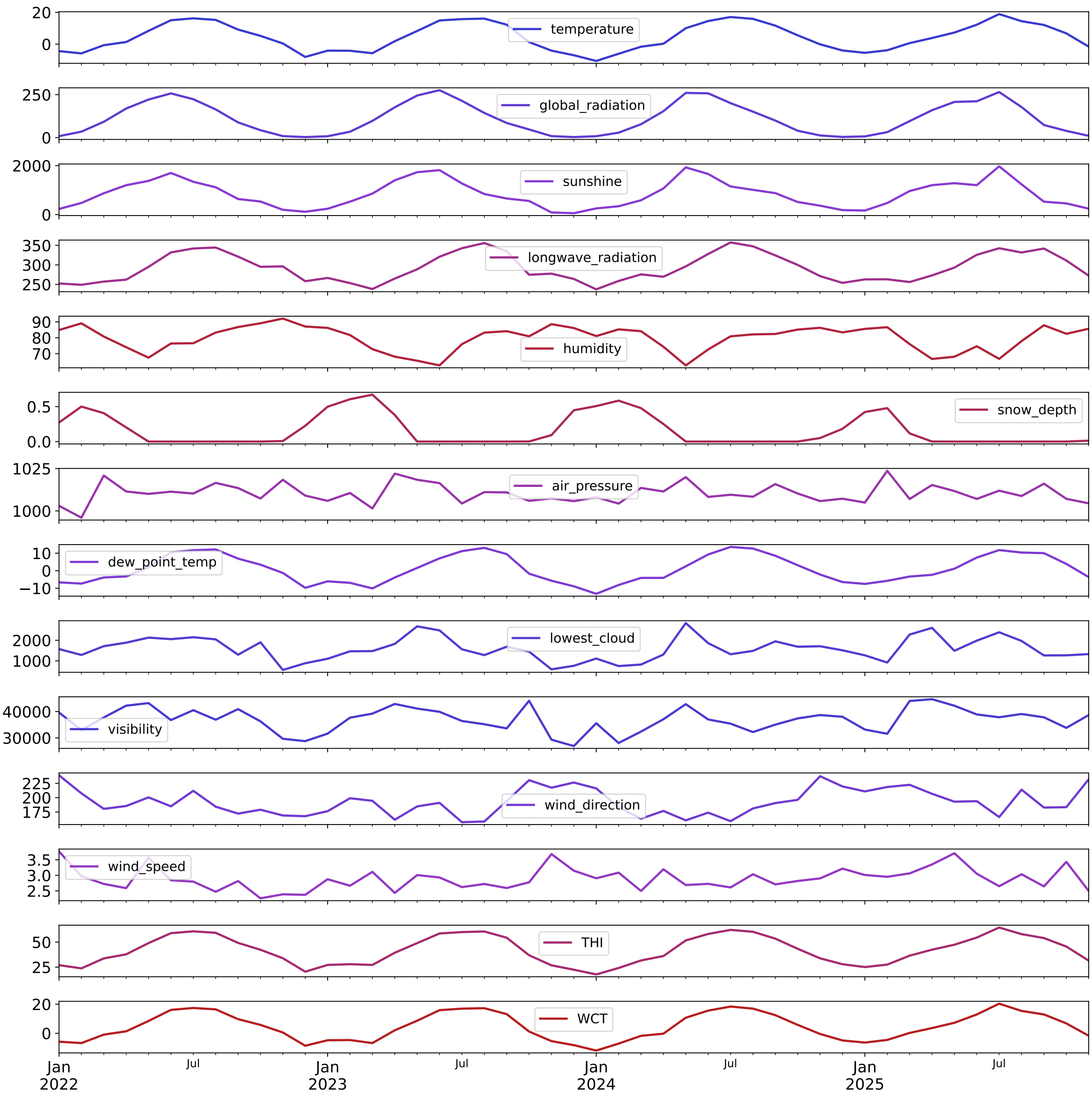

Figure 10: Weather variables' monthly values between January 2022 and November 2025.

Calendar Information

In addition to weather data, calendar information was included as external factors. During holidays, energy consumption behaviors change due to differences in social patterns. Energy consumption is not only dependent on the decision-making of individuals, it also relies on the cultural context of social and economic activities [48]. Holiday information was retrieved from Dagsmart, an open Swedish calendar information API. All holidays and weekend days during the years 2022 through 2025 were obtained, and these were compiled to a *number of holidays* feature that indicated the number of holidays for each month in the dataset. In Figure 11, the number of holidays per month is visualized, showing that December is the month with the most holidays. The fewest holidays are during the rest of the winter and autumn months. An additional *number of days* feature was constructed for each month in the dataset, since the number of days varies by month (e.g., February is missing several days of consumption compared to months with 31 days).

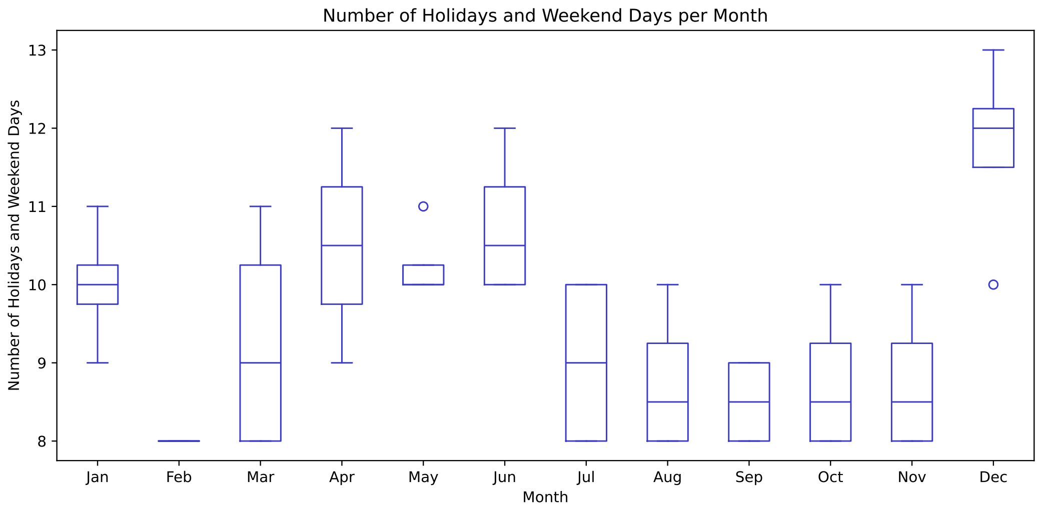

Figure 11: Number of Swedish holidays per month during the years 2022 through 2025.

3.1.3 Feature Selection

The historical consumption, weather, and calendar variables were compiled to a complete feature set for the forecasting models. A feature selection was made from the weather variables, based on the correlation maps found in Figure 9 and Appendix A. Variables with a high correlation to several other variables were excluded from the feature set, in this case global radiation, long wave radiation, dew point temperature, and visibility. This is due to highly correlated features accounting for each other's impact to the model output, and the SHAP method assuming that features are uncorrelated [38]. The complete selected feature set is found in Table 5. A subset of 2 000 random households were selected for the model training and evaluation, representing 15% of the total number of households. The data was then split into training and test data such that the years 2023 and 2024 were the training set, and January through November 2025 was the test data (see Table 6).

Table 5 Selected input features for the forecasting models.

Feature	**Description**	**Data Type**
month	Month of the year	Numeric
year	Year	Numeric
number_of_holidays	Number of holidays and weekend days during month	Numeric
number_of_days	Number of days during month	Numeric
cluster	Household cluster	Household information (categorical)
building_type	Type of building	Household information (categorical)
heating_type	Type of heating system	Household information (categorical)
contract_type	Type of contract	Household information (categorical)
temperature	Monthly average air temperature (°C)	Weather condition (numeric)
snow_depth	Monthly average snow depth (m)	Weather condition (numeric)
humidity	Monthly average relative humidity (%)	Weather condition (numeric)
wind_direction	Monthly average wind direction (°C)	Weather condition (numeric)
wind_speed	Monthly average wind speed (m/s)	Weather condition (numeric)
lowest_cloud	Monthly average lowest cloud base (m)	Weather condition (numeric)
sunshine	Monthly average sunshine amount (s)	Weather condition (numeric)
air_pressure	Monthly average air pressure (hPa)	Weather condition (numeric)
THI	Temperature humidity index	Weather index (numeric)
WCT	Wind chill temperature index	Weather index (numeric)
*lag*_1	Consumption for the previous month (kWh)	Historical consumption (numeric)
*lag*_12	Consumption for the same month last year (kWh)	Historical consumption (numeric)
*rolling_mean*_3	Average consumption for the previous three months (kWh)	Historical consumption (numeric)

Categorical features in the dataset were label encoded to integer values for all models except CatBoost. This means that each categorical value is represented by a unique integer, instead of its original string value. Label encoding, or ordinal encoding, is one of the simplest encoding techniques, and it may introduce an ordering among categorical variable values, even if one does not exist. However, for decision tree models in particular, there can be a favorable trade-off between performance and time cost by selecting label encoding [49]. For CatBoost, the original string values were preserved due to the fact that CatBoost is specifically designed to handle categorical features [50].

Table 6 Train-test split of the dataset.

Data	**Time range**	**Percentage**	**Data points**
Training data	January 2023 to December 2024	69%	48 000
Test data	January 2025 to November 2025	31%	22 000

3.2 Ensemble Learning Forecasting Models

Five decision tree-based ensemble models were selected as forecasting models; random forest (RF), gradient boosting machine (GBM), extreme gradient boosting machine (XGBoost), light gradient boosting machine (LightGBM), and categorical boosting (CatBoost). These models were selected due to their reliability, precision, and robustness in regression and classification tasks, and their ability to capture nonlinear relationships [14].

Random forest is a decision tree ensemble learning algorithm where the trees depend on a set of random feature splits at each node. This means that every tree produces its own prediction according to its set of data. The individual predictions are then combined into a final prediction via majority voting. RF is an extension of the bagging method, efficient for large datasets and robust to outliers and overfitting, making it suitable for this application. Additionally, RF is faster than other bagging or boosting techniques [51].

Gradient boosting machines are a family of ML algorithms where the learning process consecutively fits a new tree with the goal of reducing the errors of the previous one. This is done by constructing new base-learners that are maximally correlated to a decreasing loss function. The loss function can be of various kinds, which makes GBM highly flexible for different data-driven tasks. It is up to the model designer to choose the most appropriate loss function [52]. GBM is able to learn complex patterns and interactions in the data due to its iterative learning process [14].

XGBoost is a specialization of the gradient boosting algorithm with several optimizations, creating a scalable tree boosting system. The optimizations include a sparsity-aware algorithm for handling missing values, approximate learning via a weighted quantile sketch algorithm, and efficient cache access patterns reducing runtime. This results in a fast and robust boosting model using minimal resources [53].

LightGBM is another gradient boosting algorithm that focuses on efficiency for cases with many input features and when the amount of data is large. LightGBM uses two novel techniques to improve performance and efficiency, *gradient-based one-side sampling* (GOSS) and *exclusive feature bundling* (EFB). GOSS excludes data points with smaller gradients that contain less information, with the goal filtering out noisy data while retaining information gain. EFB bundles mutually exclusive features (that rarely have nonzero values at the same time) to create a smaller feature space with nearly exclusive features [54].

CatBoost is a gradient boosting algorithm that uses a technique called *ordered boosting* to reduce prediction shift caused by target leakage which occurs in other gradient boosting algorithms. CatBoost also encodes categorical features uniquely, by applying a technique called *ordered target statistics* to handle high cardinality features [50]. Because of this, CatBoost is able to achieve high accuracy, especially when the data contains a significant number of categorical features [14].

3.2.1 Configuration of Models

To configure the models, a search for the best hyperparameters was performed via exhaustive grid search. The optimization consisted of a five-fold time series cross-validation process over the hyperparameters specified in Table 7, using the scikit-learn packages GridSearchCV and TimeSeriesSplit. A subset of 30% of the training dataset was used for the grid search. For XGBoost, the grid search was minimized to 50 random combinations of the search space, in order to reduce execution time.

Table 7 Hyperparameters for decision tree ensemble models.

Model	**Hyperparameters**
RF	Trees count: 128 Features per split: auto, sqrt, log2
GBM	Iteration count: 100, 250, 500 Learning rate: 0.01, 0.05, 0.1 Depth: 5, 10 Loss type: quantile, Huber
XGBoost	Iterations: 250, 500, 1000 Learning rate: 0.01, 0.05, 0.1 Depth: 6, 8, 10 Subsampling rate: 0.5, 0.75, 1.0 Feature sample by tree/level/node: 0.5, 0.75, 1.0 Booster type: gbdt, dart
LightGBM	Iterations: 1000, 1500 Learning rate: 0.01, 0.05, 0.1 Leaves: 64 Subsample: 0.5 Feature sample by tree: 1.0 Booster type: gbdt, dart
CatBoost	Learning rate: 0.03, 0.1 Max tree depth: 4, 6, 10 L2 regularization levels: 1, 3, 5, 7, 9

All models were evaluated on the internal feature set (historical consumption only) and the total feature set (both internal and external features) separately. As a baseline comparison, an existing linear regression forecasting model developed by Umeå Energi was utilized. The model is based on historical electricity consumption and outdoor temperature.

3.3 Evaluation Metrics

To evaluate the predictive performance of the forecasting models, three statistical metrics were used; mean absolute percentage error (MAPE), root mean squared error (RMSE), and mean absolute error (MAE). MAPE is one of the most common metrics to evaluate forecasting accuracy, due to its scale-independence and interpretability. It defines the average of the absolute percentage errors, as in Equation 3.4. One disadvantage to be aware of is that MAPE yields infinite or undefined values when the actual values are zero or close to zero [55]. This thesis handles this by removing any households from the dataset that have less than 10 kWh consumption during some month. RMSE and MAE, as defined in Equation 3.5 and Equation 3.6, were utilized to give insights on the size of the errors in kWh. These two metrics are also among the most commonly used in the electricity consumption forecasting field [17].

$$MAPE = \frac{1}{n} \times \sum \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100 \tag{3.4}$$

$$RMSE = \sqrt{\frac{1}{n} \times \sum (y_t - \hat{y}_t)^2} \tag{3.5}$$

$$MAE = \frac{1}{n} \times \sum |y_t - \hat{y}_t| \tag{3.6}$$

In the equations, y_t denotes the actual observed value at time t, while $\hat{y}_t$ denotes the predicted value at time t. The total number of prediction instances is denoted by n.

3.4 Explainable AI Analysis

To enhance the interpretability of the forecasts and investigate the influence of model features, a SHAP analysis was performed for all models. The SHAP methodology consists of calculating the Shapley values for each predictor. Individual predictions are evaluated and then aggregated to create a global interpretation. This experiment used the SHAP *TreeExplainer*, specifically designed for the structure decision tree models, where the explainer traces the decision paths formed by the series of decision splits made by the models [39]. For each ensemble model, both the models' built-in feature importances and the calculated SHAP values were obtained. The results were then visualized via a feature importance plot, and a SHAP summary (beeswarm) plot displaying both feature importance ranking and the impact of individual feature values on model output. Table 8 summarizes the experiments performed to answer the research questions of this thesis.

Table 8 Summary of experiments performed to answer the research questions.

Experiment	Description	RQ
Experiment 1: Performance of forecasting models	Forecasted the monthly electricity consumption for 2000 households using five decision tree ensemble models. Performance was evaluated using MAPE, RMSE, and MAE metrics.	RQ1 – Which decision tree ensemble learning models are the most efficient for forecasting monthly residential electricity consumption?
Experiment 2: Explainable AI Analysis of Forecasts	Applied the SHAP methodology to all five ensemble models to investigate feature contributions. Traditional feature importance and SHAP summary were compared.	RQ2 – Which internal and external variables are the most important when forecasting the monthly electricity consumption of residential households using decision tree ensembles, and which variables do not benefit the forecasts?

3.5 Experimental Setup

Experiments were performed in the Python language. To process and analyze the data, the *pandas*, *seaborn*, and *matplotlib* packages were utilized. The *scikit-learn* package provided the RF and GBM models, as well as *GridSearchCV* for the exhaustive grid search. XGBoost, LightGBM, and CatBoost models were obtained from their respective GitHub repositories. The *shap* Python package was used for the explainable AI analysis.

Experiments were run in a Windows 11 virtual environment with two Intel Xeon Gold 6526Y CPU's and an Nvidia A16-2Q 2 GB GPU. For compatible models, GPU acceleration was used when performing grid search, training the models, and performing the SHAP analysis.

4 Results

This chapter presents the results to *experiment 1* and *experiment 2* performed with the aim of answering the research questions. Firstly, the performance of the forecasting models is presented, as well as a comparison with the baseline model. Secondly, results of the explainable AI analysis is presented.

4.1 Experiment 1: Performance of Forecasting Models

The performance of the five decision tree-based ensemble models was assessed via MAPE, RMSE, and MAE metrics. Table 9 shows the scores for each model, separated by internal features and total features. Scores under the internal feature category represent the models' performance when trained solely using the historical electricity consumption, while total features refers to the combination of internal and external features. The best performing model was XGBoost, with 14.96% errors on the total feature set. RF showed the worst performance with 16.33% errors and considerably higher RMSE and MAE scores on the total feature set, compared to the other models. Looking at the difference between inclusion and exclusion of external features, adding the external features results in a higher forecast accuracy in terms of MAPE. On the contrary, MAE and RMSE scores worsen for the total feature group in almost all cases.

Table 9 Performance of the ensemble forecasting models for *experiment 1*, evaluated by internal versus total features.

	Internal features			Total features		
Model	**MAPE (%)**	**RMSE (kWh)**	**MAE (kWh)**	**MAPE (%)**	**RMSE (kWh)**	**MAE (kWh)**
RF	16.34	142.47	65.93	16.33	171.47	74.68
GBM	15.67	144.07	64.49	15.07	154.97	66.67
XGBoost	15.40	144.12	63.94	14.96	153.55	65.46
LightGBM	15.67	150.21	66.09	15.18	159.54	67.03
CatBoost	15.73	152.85	65.74	16.05	150.07	66.52

A pattern occurring for all models is that on average, the models overestimate the electricity consumption during the winter months in the beginning of the year, as illustrated in Figure 12. During the summer, the average predictions lie closer to the actual average consumption of the households. The scatter plots in Figure 13 show that the overestimation of electricity consumption is a general trend for all models except XGBoost, indicated by the regression lines with larger slopes than the perfect prediction line. For those models, it is clear that predictions in the several thousand kWh range are commonly overestimated, while the errors for lower kWh predictions are more evenly distributed around the perfect prediction line.

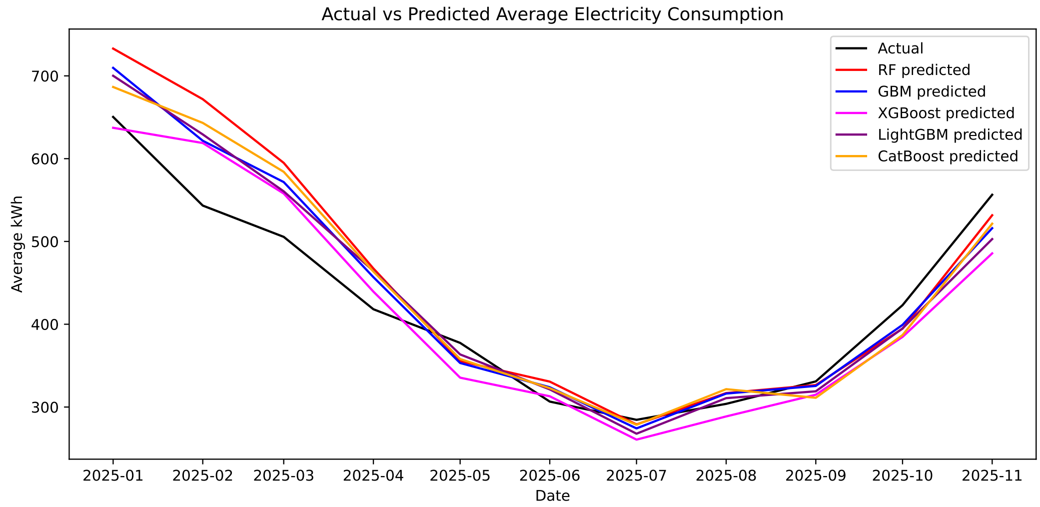

Figure 12: Graph representing the average actual versus predicted electricity consumption for all ensemble models in *experiment 1*.

4.1.1 Comparison with Baseline Model

The baseline linear regression model used for comparison utilized historical consumption and temperature as input variables. The past 24 months of electricity consumption and monthly average temperature was used to predict the following month's consumption. Evaluation of the linear regression model was performed using the same 2000 household sample as for the ensemble models, and a resulting comparison between the linear regression model and the best ensemble is found in Table 10. The linear regression model performs worse by roughly 3.5% in percentage errors. RMSE and MAE scores are also worse for the linear model, especially when compared to ensemble models using only internal features.

Table 10 Performance of the best ensemble model in *experiment 1* (total feature set) versus performance of linear regression.

Model	MAPE (%)	RMSE (kWh)	MAE (kWh)
Best ensemble (XGBoost)	14.96	153.55	65.46
Linear regression	18.47	161.43	71.98

Figure 14 shows the average predicted monthly electricity consumption for the linear regression model and all ensemble models. The linear regression model shows the same tendency to overestimate the consumption during the beginning of the year as the ensemble models do, while also showing a pattern of underestimating the consumption during the summer.

4.2 Experiment 2: Explainable AI Analysis of Forecasts

After assessing the predictive performance of all five ensemble learning models, the models' predictions were analyzed via the SHAP method. Firstly, the models were trained using

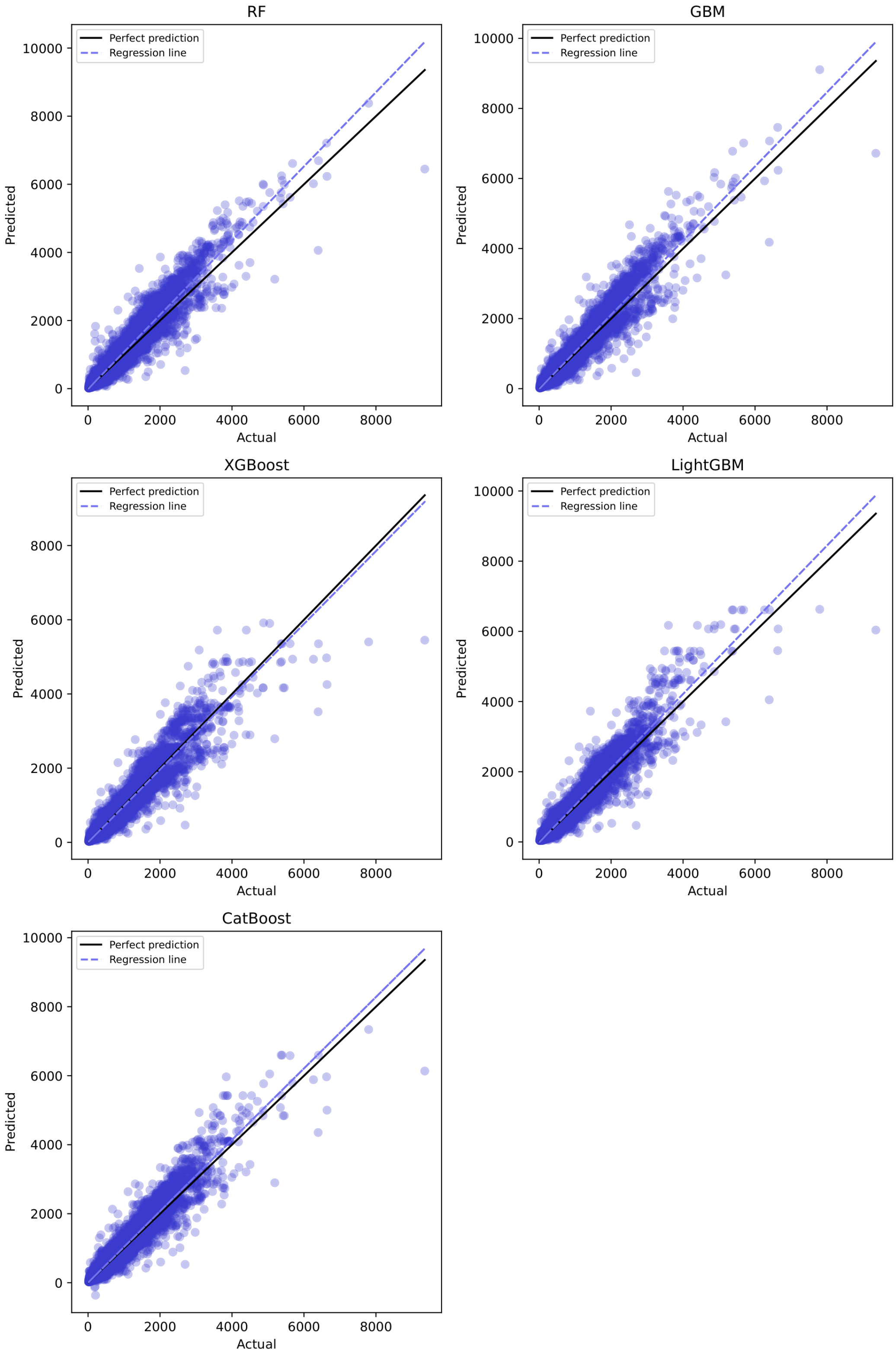

Figure 13: Scatter plots showing the prediction error distribution for each ensemble model in *experiment 1*.

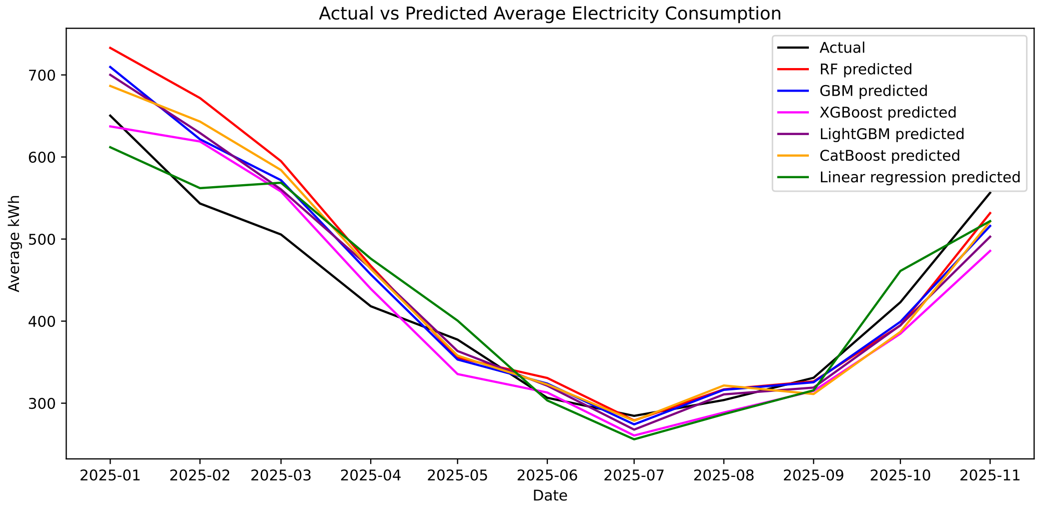

Figure 14: Graph representing the average actual versus predicted electricity consumption for ensemble models and linear regression in *experiment 1*.

their optimal hyperparameters, and feature importance metrics were retrieved for the trained models. Secondly, a tree SHAP analysis was performed to calculate the Shapley values for all features. For the RF model, approximate SHAP values[1] were used due to the full calculation being time-consuming for this particular model. Figures 15 to 19 show the resulting feature importance plots (a) and SHAP summary plots (b) for each model. In the SHAP plots, the features are ranked from top to bottom by their mean absolute SHAP values. Every instance of a variable is shown as a unique point, and the points are distributed over the x-axis according to their SHAP values. For high concentrations of SHAP values, the points are stacked vertically. The color of a point indicates whether the raw value of the particular variable instance is relatively high or low. The color distribution along the x-axis reveals the relationship between a variable's raw values and its SHAP values.

Both the feature importance and SHAP analysis show that the lag_1 and lag_12 historical consumption features are highly influential for all models. According to the feature importance plot for GBM in Figure 16, the model almost exclusively relies on the lag_1 and lag_12 features. The SHAP summary plots for all models, showing the effect of individual feature values on the prediction, also demonstrate that these features have a clear relationship where low feature values contribute to a lower predicted value, and high feature values contribute to higher predictions. Most models prioritize the lag_1 feature over lag_12 in their decision-making, except for RF. The $rolling\_mean\_3$ feature is ranked relatively high for all models, but the SHAP summary plots show that the feature may have a more complex relationship to the model output for some models. For GBM, XGBoost, and LightGBM, high values of the $rolling\_mean\_3$ feature contribute both to decreasing and increasing predicted values. The XGBoost model deviates from the other models by ranking the WCT feature the highest in feature importance. However, the more detailed SHAP summary plot ranks the lag_1 and lag_12 as most influential, displaying them with a similar pattern to the other model. The SHAP summary for this model also has a significantly lower ranking of the WCT feature.

[1] SHAP Tree Explainer: https://shap.readthedocs.io/en/latest/generated/shap.TreeExplainer.html#shap.TreeExplainer.shap_values. Retrieved 2026-05-15.

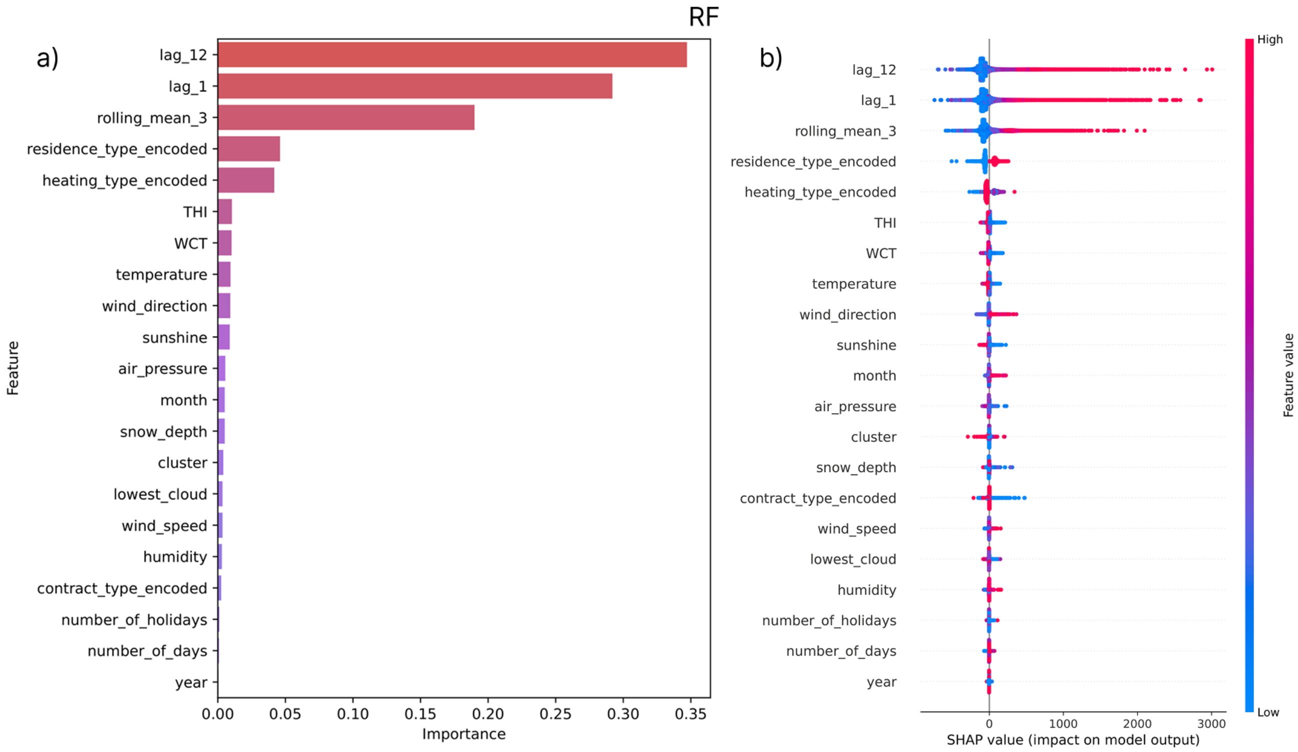

Figure 15: Feature importance plot (a) and SHAP summary (b) for the RF model in *experiment 2*.

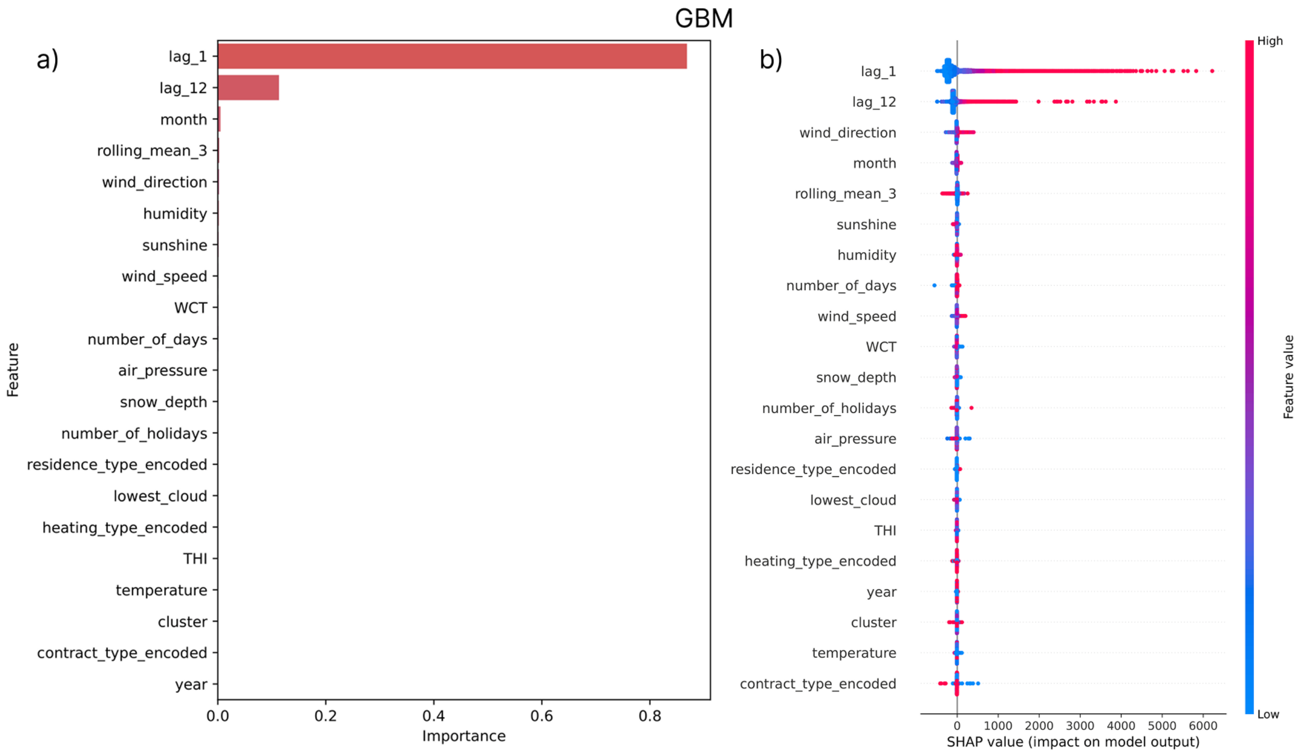

Figure 16: Feature importance plot (a) and SHAP summary (b) for the GBM model in *experiment 2*.

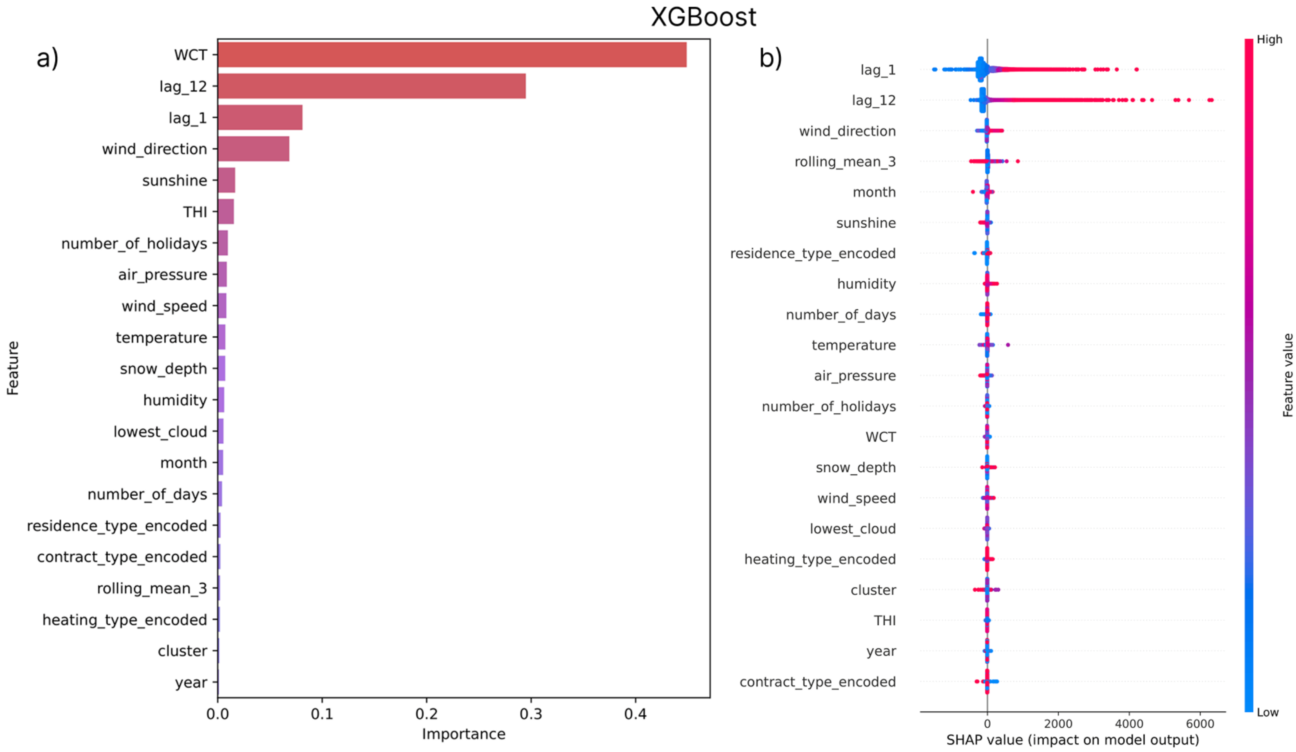

Figure 17: Feature importance plot (a) and SHAP summary (b) for the XGBoost model in *experiment 2*.

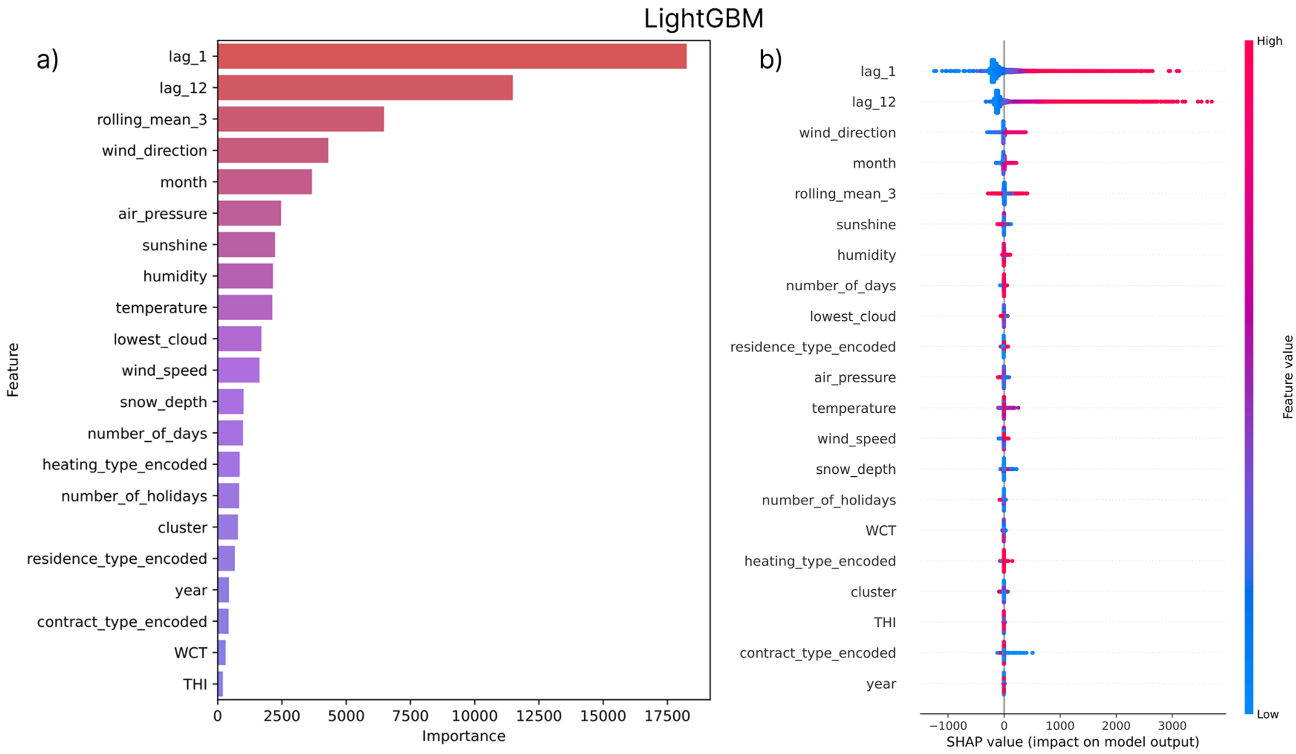

Figure 18: Feature importance plot (a) and SHAP summary (b) for the LightGBM model in *experiment 2*.

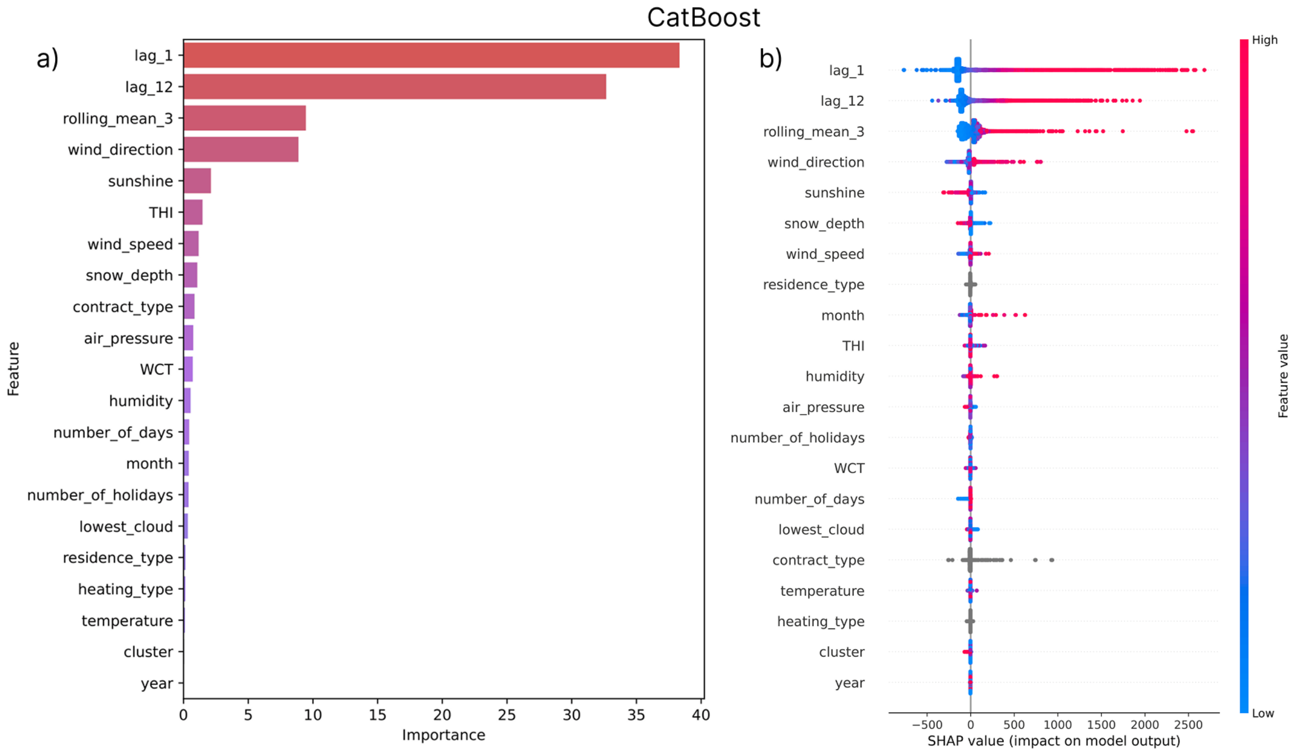

Figure 19: Feature importance plot (a) and SHAP summary (b) for the CatBoost model in *experiment 2*.

Household information features specifying residence type, and heating system were influential for the RF model, but remaining models were not as impacted by these features. The contract type feature was ranked low for all models. Even though these features commonly saw low rankings, the SHAP summaries indicate patterns where the features values had consistent impact on the SHAP values. Because of the special categorical feature handling of the CatBoost model, the SHAP summary in Figure 19 is unable to display the high/low feature value indications for these features, but it is evident that some instances of the *contract_type* feature had a large impact on the resulting SHAP value. In addition to the household information features, clustering of the households was performed based on their electricity consumption. The three resulting consumption groups (see Table 3) were encoded by the *cluster* feature. This feature was found in the lower ranks for all models.

Among the external variables, wind direction and sunshine amount were consistently ranked highly for all models. The lowest cloud base feature commonly saw a low ranking, while remaining weather variables had a more varied impact depending on the model. The calculated weather indices *THI* and *WCT* also had a varied impact on the models, which can partly be attributed to the fact that the features are fully correlated with each other and with some of the weather variables. The two calendar information features, *number_of_holidays* and *number_of_days*, are commonly ranked among the less important features for the models, along with the *year* timestamp variable. The *month* variable, indicating the month of the year, is however ranked highly for several models.

5 Discussion

This chapter interprets and discusses the results from chapter 4. Firstly, the predictive performance of the forecasting models is interpreted and further analyzed, and secondly, the contributions of individual features are assessed based the explainable AI analysis. Lastly, methodological limitations of this thesis are discussed, as well as perspectives on the implementation of the forecasting system into a real-world environment.

5.1 Performance of Forecasting Models

The forecast accuracy showed similar results across all ensemble models, with percentage errors around 15%. All models showed a large discrepancy between RMSE and MAE scores in terms of kWh, suggesting that the dataset contains large outliers. Since the RMSE metric squares the errors before averaging, large errors are penalized more, compared to MAE where all errors are treated equally via the absolute value. The considerably higher RMSE scores therefore indicate that there are some outliers with high kWh values, where the error may not be large in percentage, but the error is large in kWh due to the high consumption.

Between the internal and total feature groups, MAPE scores improved when including external features, while errors terms of RMSE and MAE increased for the total feature group. MAPE is more sensitive to errors on small consumption values, while RMSE in particular is sensitive to high consumption values and outliers. This indicates that the external factors help with increasing forecast accuracy for households with a lower and more stable consumption, while households with a high or varied consumption do not benefit from the external factors.

Household energy consumption forecasting is unique because of the stochastic nature of small-scale consumers. When forecasting the aggregated energy consumption for a collection of consumers (e.g., an entire city), the effect of randomness and user behavior is balanced out between consumers. This is not the case when performing forecasts at the individual level, since the individual customer's behavior is highly influential to the consumption [9, 16]. Additionally, this thesis utilizes a dataset with a heterogeneous group of consumers, living in various types of residence and with different kinds of energy demands, which further complicates the forecasting problem.

After obtaining the initial results, further analysis and experiments were performed to investigate the effect of outlier values on the models' performance. The data was modified by introducing a maximum monthly consumption threshold of 2000 kWh. This number was selected due to the majority of observations registering below 2000 kWh, as shown in Figure 5. Households exceeding this threshold (267 households) were removed from the sample of 2000 households used for the experiments. The grid search, training, and evaluation process was repeated for each model with the modified data, and the resulting scores are found in Table 11.

Table 11 Performance of forecasting models when outlier households (i.e., households with >2000 kWh consumption any month) are removed from the data.

Model	Internal features			Total features		
	MAPE (%)	RMSE (kWh)	MAE (kWh)	MAPE (%)	RMSE (kWh)	MAE (kWh)
RF	15.99	70.67	38.37	15.32	69.42	37.03
GBM	15.27	70.41	37.29	14.43	66.85	35.11
XGBoost	15.25	70.17	37.74	14.40	68.52	36.27
LightGBM	15.38	72.44	38.52	14.37	66.47	35.30
CatBoost	15.44	70.03	37.34	15.12	66.05	35.53

The removal of high-consuming households resulted in a small improvement in MAPE, and large improvements in both RMSE and MAE, confirming that high-consuming households have a great impact on the models' RMSE and MAE performance. The root mean squared errors were roughly half of the original for all models, and mean absolute errors decreased with around 30 kWh. MAPE scores improved up to 1 percentage point. Notably, the RMSE and MAE scores for the total feature group (both internal and external features) were better than for solely the internal group when the high-consuming households were removed from the data. This is contrary to the previous results, where the models showed better scores only using the historical consumption features. These results suggest that external variables create noise for the households with a very high consumption, and actually worsen the predictive performance for this group. On the opposite, it is clear that the external variables improve the accuracy for households with a more normal consumption, since the total feature group consistently performs better in this case.

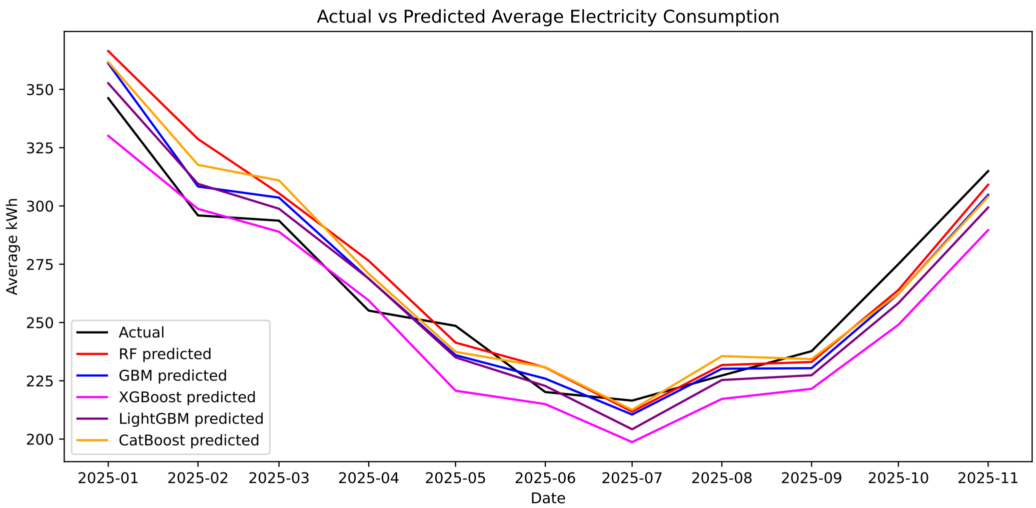

Figure 20: Average actual versus predicted electricity consumption when outlier households (i.e., households with >2000 kWh consumption any month) are removed from the data.

Figure 20 shows the adjusted plot average predicted monthly electricity consumption with the high-consuming households removed. The same pattern of overestimation in the early year still occurs, although to a lesser extent. The XGBoost model now consistently underestimates the consumption for almost the entire test period. This is confirmed by the error distribution scatter plot in Figure 21, where the regression line for XGBoost has a smaller slope than the perfect prediction line. Remaining models show a more balanced error spread where over- and underestimation is more equally distributed.

The 267 removed households almost exclusively belonged to the house category (89%) and vacation home category (10%), and the majority had electrical heating (70%). Apartment households rarely ever exceeded a monthly consumption of 2000 kWh.

Comparing the forecasting results of *experiment 1* to previous experiments can be difficult due several factors differing between experiments, particularly datasets having substantially different characteristics. However, when comparing the results to the short-term residential forecasting performed by Kong et al. [8] using LSTM, this study achieved significantly better MAPE scores (15% in this study versus 44% in their study). On the contrary, Kiprijanovska et al. [44] who also forecasted the short-term consumption for individual households received significantly better results for their deep learning framework (8% MAPE for hourly forecasts and 2% MAPE for daily forecasts). This indicates that selecting the optimal framework for forecasts is highly dependent on context, and that forecasting systems should be built with this particular context in mind.

Experiment 1 assessed the performance of five decision tree ensemble forecasting models, concluding that XGBoost achieved the best performance, although, results were similar between all models. Comparisons with a baseline linear regression model, based on historical consumption and temperature, revealed that the ensembles had a higher accuracy. These results answer research question 1.

5.2 Explainable AI Forecast Analysis

The results from the XAI analysis of the forecasts provides insights into the ensemble models' decision-making, as well as perspectives on the methodologies used. A notable observation is that the simple feature importance measure did not always rank the variables' importance in the same way as the more detailed SHAP analysis. For example, the feature importance plot for XGBoost ranked the wind-chill-temperature index WCT as the most important feature, while the SHAP summary plot showed that several other variables had a more determining impact on the model's predictions. Feature importance in machine learning is characterized by uncertainty and unreliability in some cases. The same features can result in different feature importances depending on the models, FI techniques, and subsets of data used. For non-linear models, feature importances are not fixed, but context dependent, due to variation in feature gradients. Further, the learning process for models used in this study is not the same. For random forest models, the combination och decision tree outputs happens at the end of the training process, while for gradient boosting machines, the outputs are combined at the start [56]. The utilization of SHAP analysis in addition to FI allows for a more elaborate interpretation of the variable contributions, since it reveals patterns about *how* individual features contribute to output. However, it is important to remember that SHAP results are model-dependent, affected by correlated features, and that the explanations lie in the feature rankings [38].

One pattern evident in all models is that the calendar information had little impact on the

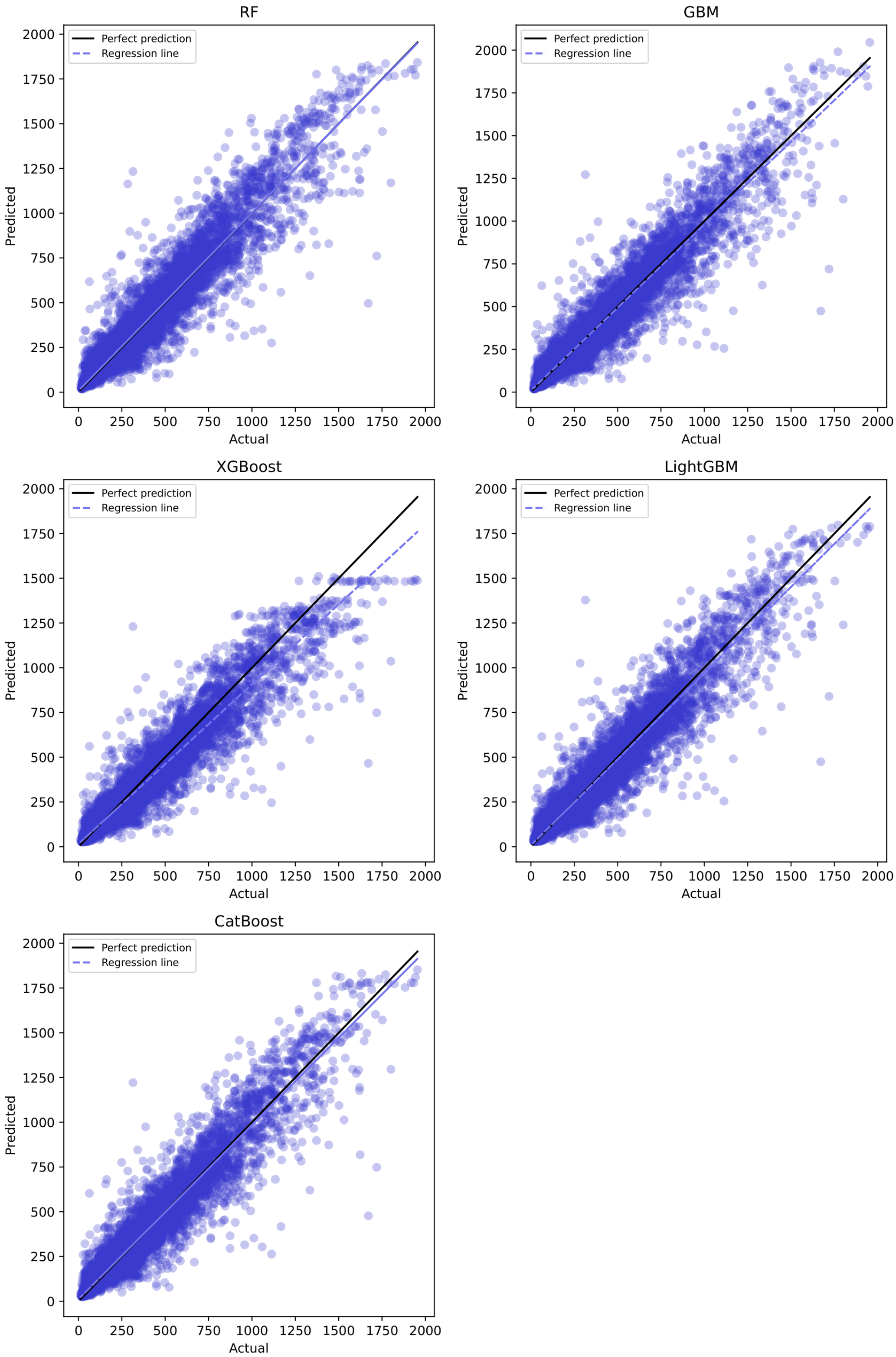

Figure 21: Scatter plots showing prediction error distribution when outlier households (i.e., households with >2000 kWh consumption any month) are removed from the data.

forecasts. Previous works have identified holiday indicators as important for electricity consumption forecasting, since consumption patterns change between weekdays and holidays [48]. Studies on short-term forecasting highlight holidays as anomalies that need to be handled in a special way, for example by introducing holidays features or by modelling holidays in a special way [14, 45, 57]. However, the results in this experiment indicate that monthly forecasts are not affected by holidays in the same way as the short-term forecasts. This could be due to the effect of holidays smoothing out over the entire month, whereas in the case of e.g. day-ahead forecasting, the entire forecast would be either for a holiday or for a weekday. It is possible that similar reasons are responsible for many of the weather variables having a weak contribution to the forecasts.

Comparing the results of *experiment 2* to Moon et al.'s work on short-term energy forecasting using XAI [14], both studies show that historical consumption is the most influential factor for the forecasts. For Moon et al., a holiday indicator feature was influential for all models, supporting the theory that calendar information is more useful in the short-term. They also found that THI and WCT were more impactful than the unprocessed weather variables, something that was not as clear in this thesis. Here, the inclusion of additional variables such as solar amount and wind direction proved to be more useful than the weather indices. Contrary to Sim et al.'s short-term forecasting study [3], this study found that time features such as year had a low influence on model output. Sim et al. found solar amount to have a weak impact on the forecasts, while this thesis found it to be one of the most contributing weather variables. These contradictions indicate that the monthly scope of forecasts results in different importances for external features compared to the short-term.

Experiment 2 enhanced the interpretability of the energy forecasts by analyzing feature contributions via SHAP and feature importance. It was found that recent consumption was the most influential for the models, and external features such as sunshine amount, wind direction, residence type, and heating system had clear patterns with the output. Other external variables showed less significant impact. These results answer research question 2.

5.3 Limitations

This thesis had some methodological limitations that may have affected the result and how it should be interpreted. Mainly, the effect of outlier values in the electricity consumption for certain households contributed to a large impact on model performance, especially in regard to RMSE and MAE metrics. Performance could potentially be improved by introducing special techniques to handle outlier households or extreme consumption values, but this experiment relied solely on input features for describing relationships in the data.

Further, this thesis is limited in the method of constructing certain features, particularly external ones. All weather variables are aggregated to monthly values using mean. Some weather variables may provide more information using another aggregation method, for example, the monthly total amount of solar radiation may be a better variable than the mean solar radiation. Additionally, the process of aggregating hourly values to a single monthly value can discard valuable information. For example, Lazzari et al. [16] found that minimum and maximum temperature were more important than mean temperature for their forecasts, possibly due to minimum and maximum temperatures having a higher variation than the mean. This indicates the existence of potential improvements by creating additional features from the weather variables than just the monthly mean. The same potential exists for the construction of more informational holiday indicators. This study only used the simple number of holidays and total number of days per month, while a better indicator could be categorizing months based on holidays.

Another limitation is the grouping of households performed via clustering. The clusters identified via the K-Means method did not show drastically different properties from each other, and the variation in electricity consumption within each cluster was high, indicated by the high standard deviation present in each cluster. Also, more than 50% of the households belonged to the high consumption profile cluster. The K-Means method may have had difficulties with grouping the data, leading to the cluster feature having a smaller impact than anticipated. More sophisticated techniques may exist to produce a grouping of households that is more informative to the forecasts. For example, Kong et al. [8] applied a density-based clustering algorithm with the ability of identifying outliers, and were able to identify four clusters based on the daily consumption patterns of 69 households. Higher accuracy may also be obtained by constructing separate models for identified clusters [44].

Some limitations were also present in the experimental execution, mostly due to computations being heavily resource-consuming for the large dataset. Firstly, the number of households used for the experiments was limited to a random subset of 15% of the total set of households. Secondly, the grid search for XGBoost was limited in order to speed up the computation, resulting in the optimal parameters in the hyperparameter space possibly not being found. Thirdly, RF used approximate Shapley values in its SHAP analysis, again to speed up computation time. The approximation does not have the same consistency as the full calculation, possibly leading to misleading variable interpretations for this model.

5.4 Real-World Application of the Forecasting System

From a real-world production perspective, the forecasting system proposed in this thesis is limited by assuming full availability of all data, which is not always the case in a real-world scenario. These experiments utilize three years of historical data electricity consumption data, as well as weather data for the upcoming month. Full historical consumption data will not be available for newer customers, creating a need for special solutions when new customers enter. Weather forecast data from open API's is available, but rarely for an entire month ahead, making the weather component of the forecasts more uncertain in a production environment.

The choice of decision-tree ensemble models in this thesis alleviates the implementation of the forecast system into a real-world scenario. Decision-tree ensembles are known for their strength in handling uncertain data with outliers or missing values [21, 22]. A robust decision-tree ensemble would therefore be suitable for this application, where some data may be missing or changing. FI and SHAP explanations showed that the models generally prioritized the previous month's consumption over the consumption for the same month last year, indicating that forecasting with these models may be possible even if the amount of historical consumption data is small.

The proposed forecasting system has potential to be implemented in a real-life production environment. The robustness and adaptability of the ensemble models is suitable for an environment where the data may be changing, and where some households may not have complete history. The addition of explainable AI analysis to enhance forecast interpretability can also be a valuable perspective for both end-users and the energy company producing the forecasts. For end-users, it helps with understanding what the forecast is based on, which in turn increases its reliability. For the energy company, the explainable AI analysis can guide the developers of the forecasting system to make informed decisions about what data to use for the forecasts. The strength with SHAP analysis is that it can be performed both locally and globally, meaning explanations can be produced for a specified month for an individual household, but also for the general decision-making of the model.

6 Conclusion

Efficient energy management is an important pillar in reducing negative impacts on the climate, especially as the total energy demand is expanding and changing. Reliable forecasts of electricity consumption can be helpful in supporting stakeholders to make more efficient decisions around energy. Firstly, this thesis performed monthly forecasts of residential electricity load using five decision tree ensemble learning models, chosen for the task due to their robustness and ability to handle diverse input features. The models were evaluated using MAPE, RMSE, and MAE metrics. Similar patterns in performance was demonstrated across all five models, but the best performance was exhibited by the XGBoost model. Compared to a baseline linear regression, the ensemble methods were able to reduce errors in all metrics, although not by an immense margin. Model performance was found to be significantly affected by the presence of extreme consumption values, largely coming from households with a specific profile. The inadequate methods of processing outliers creates opportunities for more advanced modelling, with potential to increase forecasting accuracy for various types of households.

This thesis also carried out the novel task of enhancing interpretability of monthly electricity consumption forecasts for residential households. Contrary to traditional statistical models, AI models are hard to interpret due to their black-box nature. Explainable AI analysis, which elucidates the decision-making process of models, was applied in this thesis to better understand the forecasting models and increase their reliability with users. The primary goal was to investigate a wide set of features pertaining to historical electricity consumption, household attributes, weather conditions, and holiday information, and their contributions to the forecast output. Results of the XAI analysis revealed that the previous month's consumption was the most important feature for almost all models, demonstrating the impact of recent consumption patterns. Among the external features, household attributes such as residence type and heating system showed a clear relationship to model output for most models. Weather variables with an evident relationship to the forecasts were wind direction and sunshine amount. Remaining external variables showed a varied or overall weak impact on model output. Performance-wise, it was found that the external variables increased accuracy when predicting the consumption of households with no extreme values. However, when outliers were present in the data, the impact of external variables was less clear and even reduced the predictive performance in some cases.

One of the main challenges in this thesis was processing the data and constructing input features in an optimal way to maximize model performance. The volatility of individual household energy consumption data, due to randomness in human behavior and external factors such as weather, economy, and social factors, posed a tricky challenge in modelling the problem. In addition to improved consumption pattern modelling, potential exists to handle external features in better ways, for example by using other aggregation methods than mean. Challenges were also encountered in the experimental execution due to computations being heavy for some models, creating opportunities for experimental improvements using stronger hardware.

6.1 Future Work

Future research should investigate more sophisticated techniques for handling volatility in household electricity consumption, with the goal of finding more optimal ways to model individual household consumption. This thesis applied K-means clustering to group households by their consumption patterns, but results showed that clusters had large internal variation, and that the cluster input feature had a very weak effect on model output. Better ways of handling this volatility could result in more accurate and adaptable forecasts. Ideally, a wider range of features describing the households could be included in future works to provide more detailed household descriptions, in addition to more advanced modelling.

Continuing to research forecast interpretability is also an important research direction. Explainability techniques have the potential to increase accuracy, reliability, and user-trust in forecasts, aiding stakeholders and users in their energy management. Future research should not only focus on increasing predictive performance, but also focus on developing resilient forecasting systems that prioritize interpretability and data privacy, in ways that accommodate the needs of end-users. There is also potential to perform more in-depth XAI analysis, for example by examining interactions between individual features, especially in the case where more sophisticated modelling and extra features are applied. Further work could establish outlines for turning raw XAI analysis results into easily digestible insights adapted for stakeholders and end-users, since interpreting the results requires technical knowledge of the applied techniques.

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A First Appendix

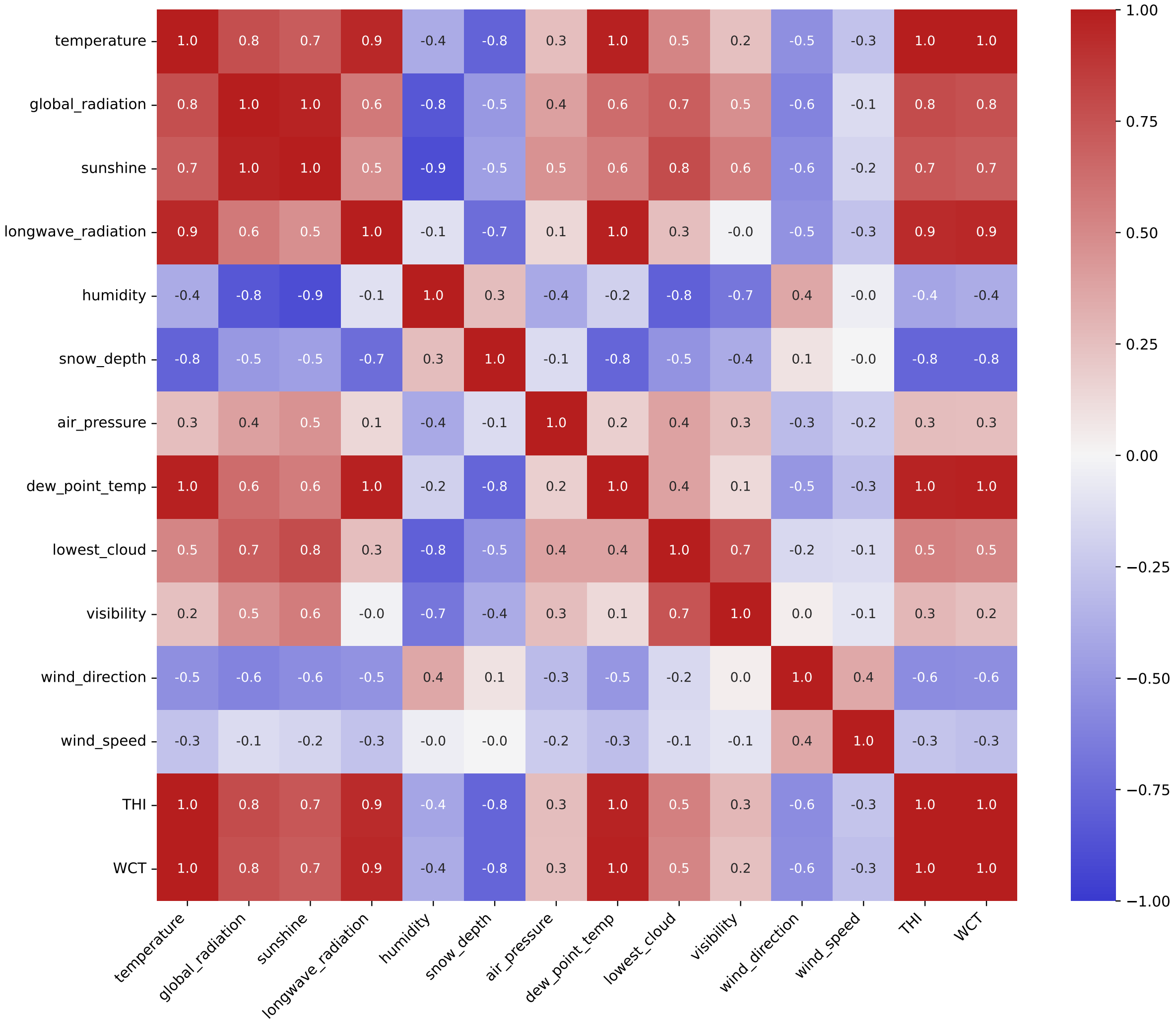

Figure 22: Correlation map for weather variables retrieved from SMHI.